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Innovating Innovation

Understanding the Role of Service Design in Service Innovation



Ana Kuštrak Korper

LINKÖPING UNIVERSITY

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by

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Abstract

Organizations across industries and sectors are seeking novel ways to innovate that can address the complex environments in which they operate. The complexity is reflected in an increasing focus on service innovation as the main growth trajectory, which entails ongoing infusion of technology into creating service offerings focused on customer experience. This creates specific challenges related to systemic and value-based dimensions of service that require organization-wide efforts in understanding and managing service innovation. Thus, in their bid to address service innovation challenges, organizations have been adopting service design as a new approach to service innovation, hoping to stir creativity by infusing the innovation process with actor-centric, systemic, and value-driven perspectives. Service design has been established in both research and practice as a human-centered, collaborative, and interdisciplinary approach that can facilitate service innovation by combining design knowledge and practices with a service perspective. However, the intersection of service design and service innovation is still in theoretical and conceptual development, making it unclear how service design might facilitate service innovation. To address this challenge, this thesis brings these two multifaceted concepts together. It employs a value-centric perspective on service innovation to articulate how service design addresses different elements of service innovation. The contributions of this thesis are based on conceptual and empirical work. The conceptual work explores the building blocks of service innovation informed by service-dominant logic and explicates meaning as a mechanism of new value creation. The empirical work uses a qualitative approach to investigate how service design facilitates service innovation. First it looks at service design projects and how they lead to service innovation through the development of new value propositions. Second it looks at service design as embedded practices that support organizational capabilities for sustained service innovation. This thesis articulates the role of service design as creative exploration and ongoing transformation and provides a conceptual integration of service and design perspectives to understand new value cocreation. Although service designers and managers are eager to implement service design in their innovation processes, they face challenges related to different, but interrelated aspects of service innovation. This research contributes with comprehensive findings that can support their decisions.

Key words: service design, service innovation, new value creation

Populärvetenskaplig sammanfattning

"Det finns ett vanligt problem med teknikdrivna företag. Eftersom de utvecklar teknik och säger; ok nu har jag utvecklat tekniken och nu ska jag sälja den. Sedan kommer de tillbaka och säger; men ingen behöver den." [Källa X]

Citatet ovan, från en av studierna i den här avhandlingen, sammanfattar en gemensam utmaning som företag som utvecklar teknikdrivna tjänster står inför idag. Deras fokus på teknik är ofta exklusiv och de saknar förståelse att deras teknik ska, vara till någons nytta, *tjäna* någon, eller något syfte. Att se teknik som en del av ett tjänstesystem tillsammans med andra organisatoriska aktörer, användare och resurser, betyder att dessa är beroende av varandra och att innovation inte uteslutande är beroende av förmågan att lösa tekniska problem. Så, metaforen om en smart individ som sitter i ett lab tills dess att ett olösligt pussel blir löst, är inte längre till hjälp för att förstå hur innovation sker, och vem som får delta i själva processen. För att en ny teknik skall antas måste den vara värdefull för användaren och det betyder att organisationer måste ändra diskurs från "Vad kan denna tekniken göra" till "Vad är det våra användare behöver". Och det är bara en början.

Idag ses innovation alltmer som ett systemiskt och värdeskapande fenomen. Ett fenomen där olika organisationer och individer, med olika, ibland även motsatta mål, behöver samarbeta i processen att skapa och anta en ny och förbättrad lösning till ett komplext problem eller ett behov som ännu inte uppfyllts. Detta innebär att, 'att utveckla' innebär att bli mer undersökande, kooperativ och användarcentrerad för att förenkla ett nytt värdeskapande för systemaktörer. I processen med att förstå denna nya verklighet, har organisationer börjat använda tjänstedesign som ett tillvägagångssätt, som ett stöd i att möta många av de utmaningar de står inför.

Tjänstedesign följer människocentrerade och systemiska principer och kombinerar dem med olika kreativa verktyg som ofta utmanar dominerande antaganden om vilka problem som bör lösas och på vilket sätt. Som sådan lyfts tjänstedesign ofta fram som ett tillvägagångssätt för innovation som kan möta dess systemiska utmaningar genom att sporra kreativt tänkande och främja deltagande och integration. Dock kan det vara besvärligt att integrera ett tillvägagångssätt som är grundat i principer som skiljer sig från de dominerande,

som till exempel fallet ofta är i teknikdrivna organisationer. Så, att förstå på vilket sätt tjänstedesign kan förenkla tjänsteinnovation för att främja ett nytt värdeskapande är viktigt. Detta är vad den här avhandlingen vill bidra till – vilken roll spelar tjänstedesign i tjänsteinnovation?

Relaterat till innovation, kan tjänstedesign ha tre olika roller: att bedriva värdecentrerad innovation, att möjliggöra hållbara innovationsmetoder, och att tillhandahålla ett tolkningsramverk för att förstå nytt värde. Den här avhandlingen visar på förutsättningar för inbäddning av tjänstedesign till förmån för tjänsteinnovation, både på en empirisk och konceptuell nivå. Den kunskapen kan hjälpa att beskriva hur tjänstedesign kan förenkla innovation, men också dess begränsningar. Till exempel spelar tajming en viktig roll om en teknikstartup vill implementera tjänstedesign för att förbättra sin innovationsförmåga. Om en etablerad teknikdriven organisation förväntar sig innovativa resultat genom att implementera tjänstedesign, finns det förutsättningar som måste övervägas, som till exempel att jämka antaganden som har att göra med definitionen av vad som betraktas som problem. Slutligen, att ifrågasätta vad det kan betyda att använda en ny teknikdriven tjänst möjliggör en mer nyanserad förståelse för hur värde kan skapas. Dessa är några av de centrala praktiska insikter som den här avhandlingen bidrar till.

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Ana Kuštrak Korper

Linköping, June 2022

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To my parents.

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1. Introduction

Change might be one of the most fundamental realities of life. Sometimes its driving force is unexpectedly fierce and empowering, other times it creeps up almost unnoticed, surprising us in all sorts of ways. And it is as much about the new things it brings as it is about our ability to respond to it. Change is also a fundamental reality of any organization, and a key constituent of any innovation. However, not every change is innovation and drawing the boundaries is not a simple task. From an organizational perspective, innovation might entail structural and strategic changes, new working principles or mindsets that lead to transformation. However, from a customer perceptive innovation should also entail changed ways of using and behaving. Where to look for sources of change and how to make changes that foster innovation? This has served as an overarching motivation at the beginning of this thesis.

1.1. Background and motivation

More than ever before, today's organizations are under imperative to innovate. Disruptive market dynamics, expansion of the service sector, competitive organizational settings and interconnected technology-infused service offerings all push organizations to find new ways to facilitate successful change in creating new offerings. Thus, organizations are increasing their efforts towards service innovation, which has become complex and multidimensional, reflecting the systemic changes in the market environment (OECD, 2016). This is happening across industries, sectors and types of organizations as many extend their offerings or transform themselves entirely to be able to support growth through service (Kowalkowski, Gebauer and Oliva 2017).

Service research has also followed this trajectory with an evolving perspective on service innovation focused on value creation. Understanding service as a basis of all exchange and viewing value cocreation as its core component are primary tenets of service-dominant logic (S-D logic) (Vargo and Lusch, 2004). As a theoretical lens, S-D logic has been instrumental in expanding the focus of service innovation to reflect the multifaceted nature of the phenomenon. Much of the focus has been on the value experienced by the customer and their active role in integrating resources to cocreate that value (Michel, Brown and Gallan, 2008). Thus, service innovation includes understanding how new forms of value

cocreation emerge through novel forms of resource integration (Lusch and Nambisan, 2015). It encompasses new offerings, processes, or business models (Ostrom et al., 2015) but is ultimately value-creating for different system actors (Edvardsson and Tronvoll, 2013). The introduction of service design into the discussion on service innovation reflects the search of many organizations for new and creative innovation approaches that can ensure sustained competitive advantage. Service design is considered to be a key approach to facilitate innovation in service systems (Patrício et al., 2018). It is often viewed as a human-centered and collaborative approach that employs designerly tools and practices in dealing with innovation (Meroni and Sangiorgi, 2016). Service design has also evolved from a stage in new service development process to a transformative driver of innovation in service systems (Holmlid, Wetter-Edman and Edvardsson, 2017, Koskela-Huotari et al., 2021) reflecting the complexity of service innovation that entails changes not only in processes, resources, or activities, but organizational conditions such as cultures and structures (Andreassen et al., 2016).

Thus, the penetration of service dominant logic (S-D logic) in service research (Vargo and Lusch, 2004) put the emphasis on the dynamic and interactive nature of value creation and broadened the research focus of service innovation. The broadened conceptualizations that moved away from the primacy of firms' output aimed to enable a more comprehensive understanding of service innovation in line with a systemic S-D logic perspective on service (Vargo and Lusch, 2016). However, this broadened perspective was also criticized for the lack of clear boundaries of what constitutes innovation, especially in terms of its outcomes (Snyder et al., 2016). Recently, Helkkula, Kowalkowski and Tronvoll, (2018) have typologized four archetypes that represent a dominant way of understanding service innovation. It includes service innovation as an outcome, process, system or experience, but archetypes are integrated in a unifying value-based service innovation framework. The emphasis on value-based service innovation is not only an attempt to reconcile its divergent conceptualizations, but it goes hand in hand with S-D logic development as a meta theory of value cocreation (Brodie, Löbler and Fehrer, 2019). Thus, value-based concerns are central for service innovation. They encompass changes in resource integrating practices thus encompassing both resource integration (Findsrud and Dehling, 2019) and value propositions (Skålén et al., 2015) as service innovation mechanisms. Therefore,

understanding the role of service design in service innovation warrants a value-based focus.

However, it is not always clear how service design contributes to service innovation. Kimbell (2011) argues that this depends not only on how we think about the service, for example by embracing or not a synthesis approach to innovation, but also how we think about design. In other words, one needs to understand what a design source of service innovation is. But the intersection of service design and service innovation where a multidisciplinary concept meets a multifaceted one can become difficult to articulate (Antons and Breidbach, 2018). Furthermore, research is currently unclear about how it sees the role of service design in service innovation. While some call for understanding it as a service development process with a design perspective (Gustafsson, Snyder and Witell, 2020), others argue that it should be included as an all-encompassing process of service system transformation (Koskela-Huotari et al., 2021). Thus, research has called for expanding service design beyond designing customer touchpoints towards understanding its transformative power in new value creation, especially through institutional change (Kurtmollaiev et al., 2018, Vink et al., 2019) and capability development (Holmlid and Malmberg, 2018, Wetter-Edman and Malmberg, 2016). Thus, understanding the connection of service design to service innovation beyond the experience level represents a research challenge.

1.2. Purpose

As the previous section shows, the broader definition of service innovation sometimes acts as an overarching concept (Carlborg, Kindström and Kowalkowski, 2014), namely for service design. This creates problems for theoretical and conceptual development of service design and service innovation (Antons and Breidbach, 2018). For example, Gustafsson et al. (2020) call for a clearer delineation between service design, new service development and service innovation. Additionally, while S-D logic as an overarching framework for service innovation used in this thesis can provide the insight into the complexity of the phenomenon, it can be difficult to operationalize it dynamic and broad elements (Witell et al., 2016). S-D logic and service design share common ground in understanding interactive aspects of value formation (Wetter-Edman et al., 2014), but theoretical integration between service and design concepts is still in development.

This thesis aims to address these challenges by delineating the space where service design and service innovation meet. Thus, the purpose of this thesis is twofold. First, this thesis aims to explicate the role of service design in facilitating service innovation. It will do so by explicating the role service design plays in new value creation related to different dimension of service innovation. Second this thesis aims to understand the service design mechanism in new value cocreation. It will do so by introducing and articulating meaning as a design source of service innovation that adds an interpretative dimension to new value cocreation.

1.3. Scope

The primary audience of this thesis are service research and service design scholars, especially those who pursue a multidisciplinary investigation of service design and service innovation. This thesis is positioned primarily within service research and service design and draws extensively on current research on service innovation and S-D logic. Apart from these, individual papers also contain conceptual and theoretical input from human-centered design and organizational capability literature, but they are not extensively discussed in the kappa. This thesis can also be of interest to design researchers and innovation management researchers, especially those investigating organizational position of design in relation to organizational capability building.

Although there has been a rising number of publications about design thinking in innovation management literature (see e.g. Carlgren, Rauth and Elmquist, 2016, Micheli et al., 2019, Verganti, Dell´Era and Swan, 2021), this thesis does not discuss similarities and differences of design thinking and service design as this is outside of its scope. This thesis considers design thinking as a management practice that is focused on problem-solving and uses design methods to inspire creative thinking among managers and other non-design professionals. While I acknowledge that service design and design thinking often converge on shared processes and tools, I posits service design as a distinct concept due to its service perspective that entails a systemic and value-based approach to service innovation.

1.4. Overview of the appended papers and thesis contribution

This section provides a brief overview of the appended paper and information about the contribution of the co-authors. The overview is presented in Table 1.

Paper 1: *Kustrak Korper, A., Holmlid, S., & Patrício, L., (2018). "Bridging design-driven and service innovation: Consonance and dissonance of meaning and value". In ServDes2018. Service Design Proof of Concept, Proceedings of the ServDes. 2018 Conference, 18-20 June, Milano, Italy (No. 150, pp. 1130-1143). Linköping University Electronic Press.*

This paper is the outcome of the conceptual study (Study 1) aimed at understanding conceptual similarities between the two major concepts in service and design research- value and meaning- and their implications for service innovation. The paper focuses on finding the common ground for understanding the concepts from a service and design perspective based on S-D logic and design-driven innovation. The paper identifies convergent conceptual dimensions of meaning and value as established concepts in S-D logic and design-driven innovation and represents a starting point for further integrative exploration of service and design research concepts.

I had a leading role in the development of the paper including reviewing the literature, theorizing, and writing the paper. My co-authors have contributed with critical feedback on conceptualization during the development and writing process.

Paper 2: *Kustrak Korper, A., Holmlid, S., & Patrício, L. (2022). "The role of meaning in service innovation: a conceptual exploration". Journal of Service Theory and Practice, 32 (2), pp. 179-198.*

This paper is the outcome of same conceptual study as the previous one (Study 1). However, this paper focuses in greater depth on understanding the role of meaning in value cocreation and service innovation. The purpose of this paper is twofold. First it is explicating meaning as the important mechanism for resource integration and value cocreation. Second, it proposes the framework for understanding the role of meaning in service innovation focusing on its role in designing new value propositions. This study contributes to service innovation

by explaining the previously assumed mechanism of how resources become by integrating knowledge from human-centered design. This opens us the space for a stronger theoretical integration of design knowledge in service innovation.
I had a leading role in the development of the paper including reviewing the literature, theorizing, and writing the paper. My co-authors have contributed with critical feedback on conceptualization during the development and writing process.

Paper 3: *Kustrak Korper, A., Patrício, L., Holmlid S., & Witell, L. (2020) "Service design as an innovation approach in technology startups: a longitudinal multiple case study". Creativity and Innovation Management, 29 (2), pp. 303-323.*

The purpose of this paper is exploring how can service design facilitate service innovation in technology startups. The paper contributes to service innovation literature by revealing how value propositions can be seen as mechanisms of service innovation and describes the key role of service design. Additionally, it contextualizes how startups might leverage service design opportunities depending on service design embeddedness.

I had the leading role in the development of this paper, including the literature review, preparation and execution of data collection and analysis, and writing the paper. My co-authors have contributed with critical feedback during data analysis and provided support in the revision process.

Paper 4: *Kustrak Korper, A., Patrício, L., Holmlid S., "We're in, now what? Integrating service design for sustained service innovation", manuscript in progress, targeted for a service journal.*

The purpose of this paper is exploring how internal embeddedness of service design practices for innovation can facilitate dynamic capabilities. This paper looks at service design as set of practices that organizations can embed to support the sensing, seizing and reconfiguring capabilities. This paper contributes to the service innovation and service design by explicating eight distinct practices that underpin dynamic capabilities and support sustained service innovation.

I had the leading role in the development of the paper, including the literature review, preparation and execution of data collection and analysis, as well as writing. My co-authors have contributed with critical feedback during data analysis and finalizing the paper.

Table 1: Summary of the appended papers

	Paper 1	**Paper 2**	**Paper 3**	**Paper 4**
	Study 1		Study 2	Study 3
Title	Bridging design-driven and service innovation: Consonance and dissonance of meaning and value	The role of meaning in service innovation: a conceptual exploration	Service design as an innovation approach in technology startups: a longitudinal multiple case study	We're in, now what? Integrating service design for sustained service innovation.
Authors	Kustrak Korper Ana, Holmlid Stefan, Patrício Lia	Kustrak Korper Ana, Holmlid Stefan, Patrício Lia	Kustrak Korper Ana, Patrício Lia, Holmlid Stefan, Witell Lars	Kustrak Korper Ana, Patrício Lia, Holmlid Stefan
Publication status	Published in Conference proceedings of ServDes2018. Service Design Proof of Concept	Published in Journal of Service Theory and Practice	Published in Creativity and Innovation Management	Working paper. Earlier version of the paper presented at the Innovation and Product Development Conference.
Role in the publication	Leading the development of	Leading the development	Leading the development	Leading the development

	the paper with critical feedback from the co-authors	of the paper with critical feedback from the co-authors	of the paper with critical feedback from the co-authors	of the paper with critical feedback from the co-authors

This thesis contributes to service innovation and service design literature in two ways. First, it articulates two distinct roles of service design in service innovation: creative exploration and ongoing transformation. While the former focuses on how service design changes value propositions in facilitating new value cocreation, the later focuses on how service design changes organizational capabilities as a key resource integrating mechanisms that enable sustained service innovation. By articulating roles of service design in service innovation this thesis addresses the problem of ambiguous conceptualizations of service design in service innovation that hinders theoretical development (Antons and Breidbach, 2018, Gustafsson et al., 2020). Second, building on the work from Krippendorff (2006) this thesis articulates meaning as a design source of value-based service innovation. Thus, it expands the interpretative dimension along the new value cocreation process and contributes to the experiential understanding of service innovation (Helkkula et al., 2018). Additionally, it strengthens the ontological and epistemological aspect of S-D logic as a value cocreation metatheory that aims to explain how and why new value emerges (Brodie et al., 2019).

Therefore, the individual papers in this thesis focus on different building blocks of service innovation. Paper 1 and Paper 2 contribute by conceptually integrating meaning and value concepts and by providing a process-based framework to explain the emergence of new value through meaning. Paper 3 and Paper 4 contribute by providing empirical support in articulating two roles of service design in service innovation.

1.5. Thesis outline

This thesis has the following structure: a kappa, consisting of chapters including Introduction, Theoretical background, Methodology, Findings, Discussion and Conclusion. Because this is a compilation thesis it also contains the papers that

are part of this thesis that are appended after the kappa chapters. In the chapters of the kappa, I discuss the following:

Chapter 1: Introduction- here I explain the background and motivation for writing this thesis. I present the problematization and discuss the thesis purpose and research questions. The introduction contains thee list and short description of appended papers and summary of the thesis contributions.

Chapter 2: Theoretical background- here I provide the literature that this thesis is based on. First, I explain the changing perspectives of service innovation. Then I focus on synthesis perspective and discuss the value cocreation as the ground for service innovation building blocks. After that I define and discuss the concept of service design and its relation to service innovation.

Chapter 3: Methodology- here I provide the overview of the methodology of this thesis. I provide a reflective description of my research journey, discuss my epistemological position and the overall research design. I provide a description of studies undertaken in this thesis with detailed information about the methodological approach in each of them.

Chapter 4: Findings- here I present my appended papers as the empirical material of this dissertation. I provide an overview of each of the four papers that contribute to this thesis discussing the purpose, findings and contributions of each.

Chapter 5: Discussion- here I discuss the findings of my thesis and relate them to the thesis purpose. I discuss the contributions for service innovation and service design and the limitations of my work.

Chapter 6: Conclusion- here I provide the final summary and conclusions of the thesis.

2. Theoretical framework

The dynamic, competitive and technology-imbued service innovation landscape emphasizes the importance of innovation that is value based and focused on customer experience. It also establishes service design as one of the most promising multidisciplinary approaches ensuring creative and human-centered solutions to ambiguous and complex problems (Buchanan, 1992, Meroni and Sangiorgi, 2016). Ever since Shostack's (1982) seminal article on designing services, service research has been recognizing the potential of service design for service innovation (Ostrom et al., 2015). Thus, an increasing body of research is bringing forward new knowledge on the intersection of service design and service innovation with new frameworks, methods, and principles (Grenha Teixeira et al., 2017, Patrício et al., 2011, Prestes Joly et al., 2019, Vink et al., 2019, Wetter-Edman et al., 2014).

However, this multidisciplinary intersection is still in early phases of theoretical development (Patrício et al., 2018). The literature calls for research exploring the theoretical integration opportunities between design and service research to establish firmer theoretical grounds for understanding the role service design plays in service innovation (Antons and Breidbach, 2018). This is the purpose of this thesis. In this chapter I will provide theoretical background for the concepts I find central to this thesis related to service innovation and service design. I will discuss the development of the service innovation concept and present the current challenges. Then I will focus on explaining service innovation as a value-based phenomenon framed within S-D logic and discuss its central tenets. Finally, I will provide the definition and overview of service design as a concept and finish with the synthesis of the ongoing research challenges related to its role in service innovation.

2.1. Understanding service innovation

Service innovation is a multifaceted phenomenon. The development of service innovation research stemmed from growing global importance of service sector and a need of technology-driven organizations to ensure competitive advantage. Thus, many organizations with innovation focus on new technology development, be it manufacturing or IT, are undergoing different variants of service transition. A common type of transition called servitization (Kowalkowski, Gebauer and

Oliva, 2017) studies manufacturing organizations shift in implementing service mindset. On the other hand, IT companies have focused on digital transformation, utilizing their technology to shift to service-centered business models (Soto Setzeke et al., 2021). Thus, the boundaries between products and services are being increasingly blurred as service transition often entails profound organizational change which has consequences for service innovation (Barrett et al., 2015).

Although the academic and practical interest in service innovation has been ongoing, recent research suggests that there is still no unifying theoretical framework to understand this phenomenon (Gustafsson et al., 2020). To some extent the reason can be traced to applying theories and methods of technology innovation to studying services and related difficulties in understanding service outcomes that were traditionally through of as fuzzier than the product or technology ones (Gallouj and Weinstein, 1997). Nevertheless, there has been a trajectory in how service innovation research has developed through three distinct perspectives. Although these perspectives coexist in service innovation research, they still represent the evolution of the concept (Coombs and Miles, 2000). These perspectives are assimilation, demarcation, and synthesis and they differ in how the boundaries and categories of service innovation are conceptualized (Carlbrog, Kindström, and Kowalkowski 2014, Coombs and Miles, 2000).

Assimilation perspective views services as same as product and thus makes no distinction between the two (Coombs and Miles, 2000). The prevailing understanding of service innovation is that technology development is its main driver thus it uses related theories explain it (Witell et al., 2016). Demarcation perspective starts from the assumption that services are different from products and thus, service innovation is different from that in manufacturing (Coombs and Miles, 2000). This perspective developed as a critique of assimilation perspective that failed to take into consideration service specific characteristics. Research appropriating demarcation perspective sees services as distinct from products, often adopting the IHIP model – intangibility, heterogeneity, inseparability, and perishability- as a primary way of distinguishing between the two (Zeithaml, Parasuraman and Berry, 1985). By putting a greater focus on service specific characteristics, this perspective approaches innovation as a relational phenomenon where customer-firm dyad is put into focus during new service development process (Edvardsson and Olson, 1996). This also entails a greater

focus on customers as sources of innovation through their involvement in innovation process (Carlborg et al., 2014) and focuses on innovation outcomes as firm-level outputs (Witell et la., 2016). However, as the boundaries between the products and services were blurring and strict separation did not make further sense (Lovelock and Gummesson, 2004), synthesis perspective, representing integration of services and manufacturing, emerged.

The assumption dominant in synthesis perspective is that service innovation should be conceptually broad enough to include both products and services. This broadened perspective focuses on value creation on a system level, taking into consideration both service characteristics and technological elements (Barrett et al., 2015). In her article about service innovation Drejer (2004) acknowledges the importance of innovation by adding value not just to organizations, but also to customers. She explains that innovation is not only product, service, organization or even industry specific, but innovation also needs to be understood from the broader perspective of affecting different systems and subsequently effecting economic development. Thus, synthesis perspective puts greater focus on customers and their changing role in the process of interactive value formation seeing them as inseparable and proactive actors in service innovation (Grönroos and Gummerus, 2014, Michel et al., 2008).

Regarding this evolution, synthesis perspective in particular has gained traction with the development and dissemination of S-D logic, that emphasizes service as the base for all value creation happening on the markets (Vargo and Lusch, 2004). In SD logic how new value emerges is of central concern for understanding service innovation. SD logic moves away from differentiating between products and services, and views service (singular) as a fundamental way of value cocreation, where products and other types of resources are vehicles for service provision and value as idiosyncratic, context-dependent and emerging from use (Edvardsson et al., 2011, Vargo and Lusch, 2008). Central tenets of SD logic, such as resource integration (Findsrud and Dehling, 2019), value propositions (Skålén et al., 2015) and service ecosystems (Koskela-Huotari et al., 2016), have all been used to conceptualize service innovation. Thus, the synthesis perspective provides good grounds to understand service innovation as a systemic and value-based phenomenon reflecting challenges many organizations are facing. The challenges of this perspective pertain to boundaries of what constitutes service innovation. Because it understands service innovation as a dynamic process it is

difficult to differentiate between the process and the outcome, and its high level of abstraction is sometimes difficult to operationalize (Witell et al., 2016). Nevertheless, synthesis perspective, opens the possibility to address the criticism of service innovation being too firm specific and without substantial focus on value it creates outside the firm boundaries (Snyder et al., 2016). Therefore, this thesis looks further into the value-based innovation, rooted primarily in the SD logic.

2.2. Value-based service innovation

Synthesis perspective developed following a lack of clear delineation between service and manufacturing sectors. It puts the focus on how new value is interactively created in multi-actor service systems, making the S-D logic a common theoretical lens in studying service innovation. S-D logic emerged a s new paradigm of markets that transcended the product - services dualism. S-D logic establishes 5 axioms and 11 fundamental premises (Vargo and Lusch, 2016) that explain how value is created on the market. The five axioms (ibid, p.18) posit that *service is fundamental basis of exchange* (1), v*alue is cocreated by multiple actors always including beneficiary (2)*, *all social and economic actors are resource integrators* (3), *value is always uniquely and phenomenologically determined by the beneficiary* (4) and *value cocreation is coordinated through actor generated institutions and institutional arrangement* (5). The axioms posit that every market exchange is a service exchange and that actors in the exchange situation apply their knowledge and skills as resources to achieve the mutual benefit from the exchange situation. Thus, service is defined as integration of resources in the value cocreation process for mutual benefit (Vargo and Lusch, 2004). This makes value cocreation to be one of the central pillars in S-D logic. In S-D logic, service providers are no longer seen as having exclusivity over value capture, with consumers being on a passive end. It acknowledges that service providers cannot fully control value creation because they cannot control how actors integrate resources or determine value. Moreover, resources are defined as either operant (such as knowledge and skills) or operand (such as tangible assets), but in both cases they are contextual and not something that is but becomes (Vargo and Lusch, 2004). Thus, service providers can only control value propositions that serve as interfaces for resource integration in new value cocreation (Chandler and Lusch, 2015).

Although service innovation can be broadly defined as new value cocreation among resource integrating actors (Lusch and Nambisan, 2015), S-D logic has viewed both value propositions and resource integration as its core mechanism. In that context Skålén et al. (2015) define value propositions as configurations of practices and resources that facilitate service innovation through their reconfigurations. On the other hand, resource integration is seen as an indispensable part of new value cocreation as it responds to changed practices leading to service innovation (Vargo, Wieland and Akaka, 2015). Thus, both value propositions and resource integration entail a social and interactive dimension of service innovation as they explain it through changes in social practices. While new value cocreation is an interactive process among multiple actors, its outcome is the idiosyncratic, contextual, experiential and meaning-laden (Vargo and Lusch, 2016). Thus, value cannot be considered as a static, but an emergent outcome that is unique for each actor and dependent not only on the interactive process (value in use) but also social context (value in social context) (Edvardsson et al., 2011). S-D logic posits that both value creation and determination are dynamic processes that can happen simultaneously (Lusch and Vargo, 2014, Vargo, Akaka and Vaughan, 2017). Although some authors critique this stance for the lack of implementation focus (Witell et al., 2016), this indicate that service innovation can be viewed as both a process and an outcome.

Recently Helkkula et al. (2018) have proposed an integrative value-based service innovation framework that addresses process, outcome, experience, and systems dimensions of service innovation, labeling them as archetypes. They argue that different dimensions of service innovation can coexist despite contradictions by emphasizing new value creation in each of the archetypes. Figure 1 shows framework of value-based service innovation.

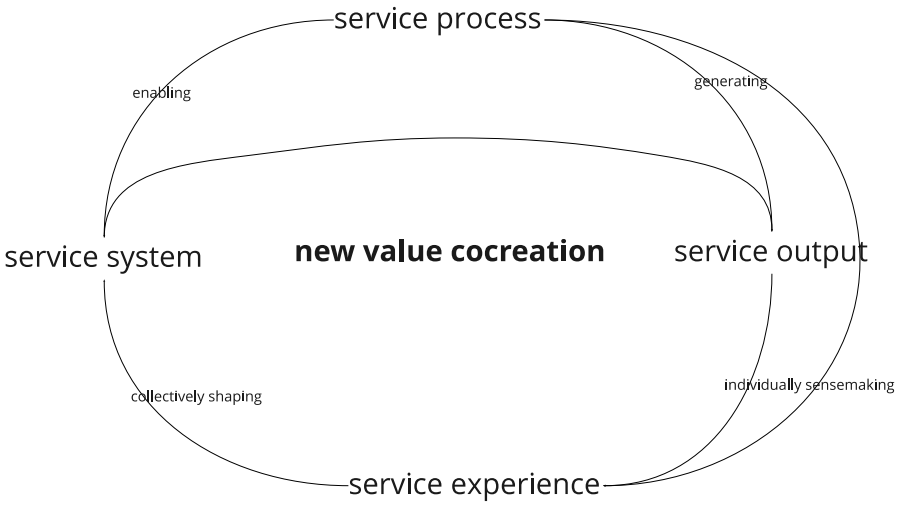


Figure 1: Value-based innovation framework, adapted from Helkkula et al. (2018, p. 293)

Value-based service innovation integrates divergent perspective and positions new value cocreation at the center of innovation efforts. In value-based innovation both resource integration and value propositions play an important role, especially in understanding experiential and systemic service innovation archetypes. Thus, while duality of process and outcome related to service innovation continues to coexist, systemic and experiential dimension provide a novel way of looking how the source of service innovation emerges. In experiential dimension, the emphasis is on individual value creating process of the customer (Helkkula, Kelleher, Philtröm, 2012) that can but does not have to happen in the joint sphere between service providers and the beneficiaries (Grönroos and Voima, 2013). In systemic dimension, service innovation is often explained through change practices that become institutionalized (Koskela-Huotari et al., 2016). Thus avoiding the dualism and focusing on the interactive value formation, service innovation can occur on individual, organizational or ecosystem level. In this thesis I will focus on service innovation on an organizational level.

2.3. Understanding service design

The intersection of service design and innovation brings forward several theoretical, conceptual, and empirical challenges that are relevant for framing the research scope of this thesis. Theoretical challenges mostly stem from the lack of unifying theories of service design in connection to service innovation, as well as its multidisciplinary perspective, which can complicate the conceptual clarity (Antons and Breidbach, 2018, Prestes Joly et al., 2019). To address these challenges this section will define service design in relation to service innovation.

In service research, service design was first used with the purpose of ensuring service quality (Gumesson, 1994). As service research increasingly focused on service quality connected to managing service front and back-end, new tools, such as service blueprint (Shostack, 1982) were developed. Visual tools aimed to capture the intangible aspects of services end ensure their consistency. Thus service design was mostly focused on designing service encounters by focusing on coordination between providers, customers, and physical surroundings (Bitner, 1990). Service design was also used as a stage in new service development (Johnson et al., 2000) as a way to redesign new services or innovate service concepts (Alam and Perry, 2002). It followed a sequential model and was introduced usually early in the development (Yu and Sangiorgi, 2014). Parallel to that, in the beginning of the 1990's service design has started developing within the design realm. Design practitioners started emphasizing experience, user-centeredness, and collaborative methods in expanding their design work to include services, especially different points of service interaction (Mager, 2009).

However, due to the multidisciplinary nature of service design concept, it is not always clear what service design entails. Kimbell (2011) suggests that service design can be understood according to the way one thinks about the service and about design. She distinguishes ways of thinking about services based on demarcation (distinction between products and services is important) and synthesis perspective (service is all encompassing concept), and ways of thinking about design as a cognitive approach (problem-solving) or an interpretative approach (design as inquiry). Thus, in conceptualizing service design according to these categories, one might differentiate between non engineering design disciplines, engineering, service engineering and designing for service. Although

I use the term service design throughout this thesis, my understanding of service design is conceptualized as designing for service.

Designing for service combines a service concept rooted in S-D logic and a design concept, rooted in several interpretative traditions of design research (Kimbell, 2011). Thus it integrates service and design perspectives that are often emphasized in discussing service design and innovation, but not always articulated. S-D logic frames the service perspective of service design as they share common conceptual ground focusing on value cocreation, resource integration, actors, and context (Wetter-Edman et al., 2014). However, in this thesis the interpretative perspective is dominantly rooted in Krippendorff's (2006) seminal work "The Semantic Turn". Krippendorff (2006) positions meaning as a core concept in human-centered design arguing that people use what they make sense of. He argues that meaning is an emergent entity of artifacts in use and that complexity of socio-material world necessitates their broader understanding. Thus Krippendorff (2006) argues that meaning of artifacts emerges in use, and spans across artifacts' life cycle, language, and ecology. Although designers cannot design nor observe meaning directly, in their design work they should proactively engage in second order understanding- defined as understanding of other peoples' understanding- to design artifacts that are flexible enough to be used based on multiple interpretations. Thus Krippendorff (2006) puts forward the concept of meaning as a new conceptual foundation of design emphasizing its interpretative dimension. So far, change in meaning has been explored as a way to radically innovate products (Norman and Verganti, 2014, Verganti, 2008). Although meaning and value seem to share common characteristics, a more detailed exploration of meaning and value as two core concepts in designing for service has not followed. However, understanding the role of meaning in service innovation can enrich the conceptual intersection of both design and service perspectives in service design.

2.4. Service design as an approach to innovation

Although there is no unifying definition of service design in relation to service innovation, service design is often defined as an approach that nurtures outside-in perspective in the design process, meaning that it primarily includes customer perspective (Holmlid and Evenson, 2008). Service design as an approach to service innovation dominantly emphasizes human-centered, creative,

collaborative and iterative principles (Blomkvist, Holmlid and Segelström, 2010, Sangiorgi and Prendiville, 2017). It emphasizes the necessity to balance between customer needs and organizational and market-based requirements in developing and implementing new service offerings (Fayard, Stigliani, and Bechky, 2017).
In order to better understand service design as an innovation approach I will briefly discuss the process and object of service design. Service design process is usually described through simple process models that involve exploration, concept development, prototyping and implementation phase (Brown, 2009). The process is iterative and diverges and converges between the phases eliciting learning and reflection. Kimbel and Blomberg (2017) articulate three perspectives of service design objects spanning across service encounters, value cocreating systems, and socio-material configurations. Service encounters represent an object that is focused on service experience through interaction with different service touchpoints. Service is experienced through engagement with touchpoints during service encounters and focus remains on provider-customer relationship. Value cocreating systems focus on competences and resources, or rather, their exchange. Here the emphasis is on actors interacting with service system. Socio-material configurations are third object of service design that are enacted in a social practice of actors involved in a service context.

Positioning service design as an innovation approach has resulted in extensive research contributions related to method development under the theoretical umbrella of S-D logic (Clatworthy, 2011, Grenha Teixeira et al., 2017, 2012, Patrício et al., 2011) and mostly focused on services at the level of interaction (Secomandi and Snelders, 2011). However, recent research has problematized the narrow scope of service design and proposed a broader agenda (Vink et al., 2021). Koskela-Huotari et al. (2021) argue that service design should be considered as a practice that facilitates service system transformations. Building on S-D logic and institutional theory, they argue that service design can benefit from research that investigates service design beyond periodic projects and looks at service innovation as a systemic phenomenon. In line with this, recent research has investigated organizational aspects of service design integration. For example, Karpen, Gemser and Calabretta (2017) provide a framework of organizational conditions for service design integration consisting of individual level abilities, interactive practices, and organizational capabilities. Their multilevel framework emphasizes the opportunities to building organizational capabilities for innovation through service design. Similarly, Wetter-Edman and Malmberg

(2016) have focused on organizational ability to absorb external knowledge as a way of building design capability. They have concluded that service design project can support explorative learning, but they have limited impact on exploitative learning. Thus embedding service design in organizations to support ongoing service innovation is complex (Bailey, 2012) and dependent on organizational structures and design legacies (Junginger, 2015).

Service design as a multidisciplinary concept builds on a synthesis perspective of service innovation and interpretative design dimension. Although research in service design has moved beyond NSD (Holmlid et al., 2017) service innovation research often views service design as a development process with a design perspective (Gustafsson et al., 2020) while not articulating what design perspective entails. More recent research in service design suggested that service design should broaden the scope of inquiry and engage with service systems and social structures as materials of design (Vink et al., 2021). This also entails a more detailed understanding of service design capabilities (Holmlid and Malmberg, 2018, Wetter-Edman and Malmberg, 2016), and organizational conditions that enable or hinder its embeddedness (Bailey, 2012, Junginger, 2015). Research on the intersection of service design and service innovation often expresses different goals that can hinder its conceptual and theoretical development (Antons and Breidbach, 2018). Therefore, this thesis articulates the role of service design in service innovation.

3. Methodology

This chapter discusses the research approach and the methodology of this thesis. I will first reflect on my research journey to provide the necessary background for understanding the emergence of my conceptual and empirical studies. Then I will discuss the context of my studies followed by my research approach and my ontological and epistemological positions. Finally, I will give an overview of my studies and finish with an evaluation and critical reflection on the research work.

3.1. Research journey

The contributions of this dissertation are distributed over three studies, one conceptual (resulting in two papers) and two empirical (resulting in two papers). Such distribution of conceptual and empirical work requires further clarification. Although I regard conceptual work as framing the empirical work on a broader level, these are loose connections, meaning that the proposed conceptual frameworks have not been directly tested via empirical studies. This is due to multiple idiosyncrasies in my research journey as well as the initial decision to do a compilation thesis instead of a monography. Thus, to clarify the connection between the conceptual and empirical work in this dissertation, and to provide the context behind it, I will first reflect on my research journey.

I started my PhD within the Service design for innovation network (SDIN)[1], an EU-funded training network that was formed to educate a new generation of researchers and develop multidisciplinary research contributions on the intersection of service design and innovation. With an educational background in business administration, I was primarily motivated to do research about service design to understand the implications of this novel (to me) practice in dealing with innovation challenges in services management and marketing. Additionally, the research project tied to my SDIN position was focused on service design for innovation in the context of emerging technologies. My initial research work was guided both by my motivation to do research on service design as an innovation approach and by my research project, which was more specifically oriented toward studying the technology context. I became immersed in literature on service research, innovation management and design, which reflected my early

[1] My PhD position within SDIN was first with the University of Porto (2016-2018) and then with Linköping University (2018-now)

attempts to make sense of the multidisciplinary nature of the service design concept (Prestes Joly et al., 2019). Early on in my research, service-dominant logic (S-D logic) provided a theoretical lens through which to understand service innovation as a value-based and systemic phenomenon (Lusch and Nambisan, 2015, Vargo and Lusch, 2016). This position was also reflected in service design literature (Holmlid and Evenson, 2008, Maffei et al., 2005, Wetter-Edman et al., 2014). On the other hand, apart from the design-driven innovation framework (Verganti, 2008), which used the concept of meaning from human-centered design (Krippendorff, 2006) to explain radical product innovation, much of the innovation literature tied to technology seemed to lack conceptual convergences that could support value-based and systemic innovation perspectives. However, exposure to the concept of meaning as a design concept used in innovation management literature triggered my curiosity. I was intrigued by many similarities between the concepts of meaning and value that S-D logic did not discuss directly, but that seemed relevant in establishing the conceptual position of service design as an innovation approach.

From these early musings, and a couple of months into my PhD, the idea arose that ultimately became embedded in my conceptual study: how similar are the concepts of value in service research and meaning in design research, and what theoretical implications could this have for understanding service innovation? Around the time I started to explore these concepts, which would later result in an integrative conceptual framework, an opportunity to empirically study service design as an innovation approach in technology startups arose. It was an opportunity to observe and explore how service design can facilitate innovation of technology-enabled services in startups through a longitudinal exploratory study, where service design was introduced through innovation projects. Technology startups are an important part of an innovation ecosystem, but research on how service design might facilitate innovation in this organizational setting was scarce, and therefore this empirical project was meant to address this particular research challenge. The longitudinal case study enabled an ongoing observation of service design introduction into five Portuguese technology startups without prior service design experience. It also provided insight into opportunities and challenges related to designing new value propositions.

Thus, during my first empirical study, the main body of my conceptual study was emerging in parallel, which in the end resulted in two published papers (Kustrak

Korper et al., 2018, Kustrak Korper et al., 2021). However, developing and publishing conceptual work proved to be challenging. By the time the first empirical study was finished, only a smaller part of the conceptual study had been published, while the majority of what I considered to be the main contribution was under ongoing revision. During that time and given the experience of working with conceptual study, my research efforts were redirected toward designing the second empirical study. Following the findings from the study with startups, which described how service design can facilitate new value propositions and ways in which startups might embed service design as an innovation approach, I wanted to further investigate how embeddedness of service design could support service innovation in technology-driven organizations. Due to the access granted by the SDIN project to one such organization in Portugal with established internal design practices, I had an opportunity to learn how this experience has transformed the development processes but also built the organization's capability to work with design. Although there are studies advocating for the transformative role of service design (Junginger et al., 2015, Koskela-Huotari et al., 2021, Kurtmollaiev, et al., 2018), there is a need for further empirical evidence to understand how it might develop organizational capabilities that enable transformation. Therefore, I decided that the focus of the second study should be on studying the role of service design in creating organizational capabilities that support service innovation. The second study focused on understanding service design embeddedness in established technology-driven organizations that have been using service design internally for at least four years at the time of data collection. I was able to reach informants in design roles across nine established technology-driven organizations from various industries. The study provided insights into the practices that are important for developing design-related organizational capabilities.

Although I will elaborate further on the research approach and data collection methods of all the studies in the following sections, this section's aim is to provide background information about how the studies emerged to understand the ways they connect. Thus, what connects the studies is their focus on service design as an enabler of service innovation in a technology-driven context. S-D logic frames the definition of service innovation in this dissertation, while individual studies explore its different building blocks. In the following paragraphs I will further discuss my research approach.

3.2. Research approach

The purpose of doing any research is to advance knowledge, and a research approach is one necessary element in understanding how this knowledge came to be. Through research design and methods, the research approach also reflects a philosophical worldview that influences how a researcher goes about their research practice (Creswell, 2014). This thesis aims to understand how service design facilitates service innovation in technology-driven contexts. As discussed in Chapter 2, both service innovation and service design are phenomena that have not yet converged into theoretical maturity (Antons and Breidbach 2018, Sangiorgi and Prendiville, 2014, Snyder et al., 2016). Thus, this research takes an exploratory stance and uses a qualitative research approach.

The choice to do qualitative, exploratory research co-evolved with my worldview that people make sense of the world based on their own experience situated in a social system. Thus, my research is positioned within social constructionism, a worldview that takes the position that knowledge and understanding are socially situated and constructed from interactive experiences and reflections about them (Creswell, 2014). Although social constructionism is a fragmented philosophical position (Alvesson and Sköldberg, 2009), in my research approach I use its main defining tenets: it relies on the understanding that the knowledge and concepts we study are constructed rather than discovered, but that they are connected to real entities (Berger and Luckman, 1991). Thus, it takes a subjective epistemological position. This paradigm assumes that people create subjective meanings out of their situated experience, and the role of a researcher is to work with the complexity of these meanings, often attained through open-ended inquiry with the goal of building theory (Creswell, 2014). While theory building is considered to be a primary purpose of a qualitative approach, Graebner et al. (2012) caution that this can often be mischaracterized, restricting the other goals and possibilities of working with qualitative data. They argue that providing a clear rationale behind any qualitative approach may include not only building but extending the theory, understanding process issues, illustrating abstract ideas, examining linguistic phenomena or some combination of those. This applies to this thesis as well, where my approach serves to extend the current theoretical knowledge and illustrate abstract concepts related to service design and service innovation from the S-D logic domain that I use as a theoretical lens.

As a researcher I have assumed the role of co-constructionist of knowledge. I have attempted to be aware and reflexive about my own personal and social experiences influencing the process of the qualitative inquiry I have undertaken. My research approach has relied on theorizing with concepts and empirical material aimed at unlocking and clarifying different aspects of service design and its connection to service innovation. Theorizing is a process that "consists of activities like abstracting, generalizing, relating, selecting, explaining, synthesizing, and idealizing", which generates outputs that serve as placeholders for theory generation. (Weick, 1995, p.389). This interaction between the conceptual and empirical work has been continuous. I have engaged in abductive reasoning to make sense of my empirical material and its connection to theory (Alvesson and Sköldberg, 2009). Abductive reasoning starts with induction but acknowledges that interpretation of empirical material is inevitably informed by existing knowledge of concepts and theories. Although getting immersed in the data was a starting point for my research, my own understanding of it also co-evolved with my knowledge and reflection about concepts and theories I used during the research process. This applies to both conceptual and empirical studies. In the following sub-sections I give an overview of my studies with information about data collection and analysis.

3.3. The technology-driven context

This thesis explores service design as an innovation approach that is situated in a technology-driven context. In this thesis, I define the idea of a technology-driven context not as industry-specific, but rather as encompassing organizations with value propositions that have a strong technology component, innovation competence focused on research and development (R&D) and organizational cultures that support it. This means that the context covers science-driven (e.g., chemicals), high-tech (e.g., IT) and dominant design (e.g., car manufacturing) industries (Boutellier and Heinezen, 2014), as well as organizations of varied sizes and maturity levels. This context was of particular interest to me because of its richness and innovation focus. Specifically, there is an increasing number of technology-driven organizations that view design as a key to innovation and are trying to legitimize design as an organizational function (Micheli et al., 2018). Industry specific reports reveal that an increasing number of technology companies are not only repositioning themselves and their offerings as services but are also increasingly using design methods and tools to successfully innovate

across organizations (Forrester Research, 2018). Design, of course, is not an unfamiliar organizational function in many technology-driven organizations, from car manufacturers to software developers, although it has traditionally been confined to a specific stage in the product development process (Holmlid, Wetter-Edman and Edvardsson, 2017). On the other hand, in order to support innovation across an organization, design needs to be viewed as a core organizational activity that accompanies the evolving dynamic of change (Junginger, 2015, Sangiorgi et al., 2019). This often entails reshaping the idea of technology as a product towards technology as a value proposition, and also requires a mindset shift in understanding the concepts of service, design, and innovation (New, 2008). Additionally, service design studies often focus on organizations in the public sector and healthcare in explaining value-driven and systemic challenges of service innovation. (Malmberg and Wetter-Edman, 2016, Patrício et al., 2018, Vink et al., 2019), although there are studies investigating the introduction of service design into different technology-driven contexts (Costa et al., 2018). However, in the context of technology-driven organizations, engineering culture (Schein, 1996) is often dominant and might hinder learning opportunities that come with novel approaches (Ford et al., 2000), which has consequences for innovation outcomes. Additionally, as technology becomes ubiquitous and enables new forms of interaction, the boundary between products and services is becoming less distinct (Carlborg et al., 2014). This is reflected in the ongoing transformation of many technology-driven organizations. Such transformation requires the development of practices and capabilities that support the customer experience and can address systemic innovation challenges (Zaki, 2019). Therefore, understanding the role of service design for innovation in contexts where engineering approaches have traditionally had primacy can enable organizations to make sense of the alignments between their innovation efforts and the design work that supports them.

Regarding the studies in this thesis, both empirical studies are situated in technology-driven contexts, while the conceptual study uses illustrative examples from the same context. In the first empirical study, the technology-driven context includes technology startups, where service design as an innovation approach was introduced through a service innovation project. In the second empirical study it includes informants in design roles with established technology-driven companies with internal service design practices. Technology startups and established companies offer a rich and varied research context for exploring service design.

For technology startups this has implications with regard to understanding how service design supports or hinders the innovation process in various stages of the startup life cycle, while in established organizations it has implications for understanding the role of service design in establishing organizational capabilities for sustained innovation.

3.4. Overview of the studies

This thesis consists of three studies that resulted in four papers. I will first discuss the conceptual study that yielded Paper 1 and Paper 2. Next, I will discuss the empirical studies. The longitudinal multiple case study resulted in Paper 3 and the qualitative interview study resulted in Paper 4. An overview of the studies is presented in Table 2.

Table 2: Overview of the studies

Study no.	Type	Purpose	Approach	Empirical material	Contributing papers
1	Conceptual	Role of meaning in value cocreation and service innovation	Problematization (Alvesson and Sandberg, 2011)	Focal literature review	Paper 1 Paper 2
2	Empirical qualitative	Service design approach to designing new value propositions	Longitudinal multiple case study (Yin, 2014)	Semi-structured interviews (14), observation notes, project documentation, other archival data	Paper 3
3	Empirical qualitative	The role of service design in building	Qualitative interview study (Miles and	Semi-structured interviews (21), project	Paper 4

		organizational capabilities	Huberman, 1984)	documentation, other archival data	

3.4.1. Study 1 (conceptual)

The purpose of Study 1 was to understand the role of meaning, as a human-centered design concept, in value cocreation and service innovation. Conceptualization involves conceptual thinking which is "the process of understanding [a] situation or problem abstractly by identifying patterns or connections and key underlying properties" (MacInnis, 2011, p.140). Although many empirical studies involve some level of conceptualization, this process is the core of all conceptual studies. Conceptual studies build theories and often do not contain any empirical insights, which can hinder their clarity, especially their "methodological" aspect (Jaakola, 2020). Thus, it is important for conceptual studies to communicate the choice of focal concepts and theories they use in the process of building conceptual contributions, and to explain the motivation behind these choices. The purpose of Study 1 was to question the assumptions underlying the value cocreation process defined in S-D logic (Vargo and Lusch, 2016). S-D logic explains the mechanisms of what is happening during value cocreation but remains generic about what drives the engagement of actors in value cocreation for the actors' mutual benefit (Vargo and Lusch, 2004). This type of approach, which relies on questioning assumptions, is considered necessary in building relevant theoretical contributions. Alvesson and Sandberg (2011) define this approach as problematization and view it as a paradigm-free methodology that rests on unpacking one's own familiar theoretical position to open the space to interesting research questions. Study 1 relied on this methodological approach. It used S-D logic as a focal theory (Jaakkola, 2020) and argued that the mechanisms underlying the resource integration and value co-creation processes are assumed and not explicit. Without understanding why these processes ensue, service providers lack the comprehensive perspective that is necessary for designing meaningful value propositions, which can have implications for the outcome of their service innovation efforts.

Consistent with the problematization approach, the study used selected and relevant literature from S-D logic, design research and innovation management as

research material based on which conceptual frameworks are built. Mainly, it introduced the concept of meaning from human-centered design (Krippendorff, 1989) as a supplementary concept (Jaakkola, 2020) to clarify and explicate the value cocreation process and suggest interpretative mechanisms behind resource integration. The focal literature was selected based on its relevance for explaining the concepts of value and meaning. The focal literature selection process started with the focal literature domain of S-D logic. It included journal articles and book chapters, which were predominantly conceptual. Also, literature in human-centered design that explained the concept of meaning was selected and complemented with design-driven innovation literature. Other relevant papers were selected based on a snowball-like literature sampling process. This provided data for analytical work that resulted in an integrative framework (MacInnis, 2011). The type of theorizing in this study most closely resembles the narrative style (Cornelissen, 2017). However, one of the limitations is that narrative-style theorizing can often come across as too descriptive and vague. To address these limitations, I have used visual representations as process models explicating meaning as a prerequisite for value cocreation and its role in service innovation. Additionally, this study uses illustrative empirical examples to support its claims and to make abstract concepts more accessible.

3.4.2. Study 2 (empirical)

The purpose of Study 2 was to understand how service design as an innovation approach might facilitate service innovation in technology startups. It uses a case study approach that has been common in studying various organizationally-related issues (Eisenhardt, 1989). The use of case studies can be defined as a methodological approach that seeks an in-depth understanding of a phenomenon in a real-life context, often using multiple data sources to get rich descriptions based on which interpretations are made (Yin, 2014). It is typically concerned with answering the how and why questions that are connected to a specific (or multiple) bounded system(s); each system is a case (Creswell, 2014). Additionally, case studies are often advocated as an approach to theory generation when the phenomenon under study is not well understood (Eisenhardt, 1989). While most case study methodology authors will agree on these determinants, there are different ways in which case studies are conducted, especially the extent to which the process is prescribed (Creswell, 2014). This is related to one of the goals of the case study approach, which influences the type of data that is collected

and how it is analyzed and interpreted. For example, Yin (2014) argues that the use of case studies as a methodology is bound by specific requirements and procedures entailing systematic and rigorous data collection and analysis protocols that can serve both deductive and inductive analysis. Although openness to interpretation remains one of the main tenets of a case study approach, some argue that its highly structured procedures can be restrictive and more appropriate to a positivist research paradigm (Denzin and Lincoln, 2008). This study followed a structured procedure of planning, data collection, and analysis closer to Yin's (2014) recommendations.

Study 2 was designed as a longitudinal multiple case study. Multiple case design enables collection of more compelling evidence since it relies on replication logic (Yin, 2014, p.57), while longitudinal studies enable rich insights and can enable inference of potential cause and effect relationships (Leonard-Barton, 1990). The case study protocol was developed at the beginning of the study and prescribed what types of data would be collected and when they would be collected. The study was planned to cover technology startups participating in service innovation projects that were focused on leveraging their existing technology through service design, as well as post-project outcomes. As the study spanned a period of 11 months and I had planned for several points of data collection during that time, it was important that the cases not only wanted to participate in the project but could also commit to the longitudinal nature of the study. The final case selection included five technology startups that had no prior experience with service design. A major part of the study was structured around service design projects conducted in collaboration with master students during a service design course. However, the cases were focused on startups, and therefore on startup founders, who were active participants in the projects and the main sources of information throughout the study. The study was, thus, designed to enable not only understanding of how the service design project facilitated service innovation, but also whether service design permeated innovation processes in the startups after the projects finished.
The data collection occurred over a period of 11 months: prior to the start of the project, during the project and after the project finished. Before the project started, data was collected to understand the context of each startup's founding, current innovation processes and detailed operational circumstances. During the project, data collection involved observations as the project evolved. After the project finished, data collection was focused on understanding whether service design had become embedded in the companies' innovation processes and if so, why this

was true. The data for Study 2 consisted of 14 semi-structured interviews, archival data, and observations to ensure triangulation (Yin, 2014). Interviews were done based on an interview protocol and were conducted in three instances: before the project began, 2 months after the project finished (1st follow-up) and 6 months after the project finished (2nd follow-up). In the second follow-up, one of the startups was facing operational challenges and was unavailable for the final interview. All interviews were conducted in English and transcribed. While semi-structured interviews were used before and after the project, participant observation was the main form of data collection during the project as it enabled me to directly follow the process of working with service design as the projects progressed. This included participating in meetings and codesign sessions with students and startup founders, as well as kick-off meetings, final presentations and feedback sessions. During these observations I used notetaking to collect insights, while the project documentation, which included different visualizations across different stages, served as supporting material.

Data analysis was performed using an abductive approach (Alvesson and Sköldberg, 2009). I started my analysis by looking at the data as I collected it, but I did not used predefined categories in my analysis. Also, my analysis was informed by conceptual and theoretical insights about service innovation and service design. After initial coding, there were 2 major iterations of analysis and discussion among co-authors, concerning both in-case and cross case analysis, until themes became established.

3.4.3. Study 3 (empirical)

The aim of Study 3 was to investigate how the internal embeddedness of service design supports organizational capabilities for sustained service innovation. In this study, the focus was on expert service designers in established technology-driven organizations. This focus was chosen based on current research challenges that are concerned with understanding the complexity behind embedding service design in organizations (Holmlid and Malmberg, 2018, Junginger, 2015) and the role that design plays in supporting organizations' efforts to remain innovative (Wetter-Edman and Malmberg, 2016). This study was designed as a qualitative interview study (Miles, Huberman and Saldaña, 2018) aimed at understanding expert service designers' practices underpinning organizational capabilities for sustained service innovation. Thus, it was important to include expert designers working in organizations that proactively and purposefully used service design in

their internal innovation work. This focus on technology-driven organizations with ongoing service design practices enabled a more in-depth exploration of service design role in capability building that would have been more difficult if the focus had been on occasional service design projects with external experts. Since this study is situated in a technology-driven context it was important to have access to informants working in different industries and at different levels (excluding only the entry level due to potential lack of experience and insights). This enabled variability in experiences and perspectives. As it was important to reach informants who had been working with service design in technology-driven organizations, this study uses theoretical sampling (Eisenhardt and Graebner, 2007). I started with a list of organizations known to use service design and reached out to informants who might be willing to participate in the study.

The main instrument of data collection was semi-structured interviews based on an interview protocol. Interviews are a common data collection method in qualitative studies that elicit detailed information in the form of social interaction between the interviewer and interviewee, where knowledge is jointly constructed (Brinkmann and Kvale, 2015). The interview protocol used in this study served as a guide to address major topics of interest about the phenomenon while still allowing informants to share their experiences in a way and that was meaningful to them. The focus was on understanding the main tenets of service design practice and the organizational conditions that support or hinder it. All informants were informed about the purpose of the study in advance, and prior to the data collection were presented with informed consent documents, which they signed. The study contains 21 interviews in total with informants across nine technology-driven organizations that are considered innovative in their respective industries. Interviews, lasting between 50-90 minutes, were conducted in person and over the phone over a period of 19 months. The interviews were conducted in English and all but one was recorded and transcribed verbatim. For the one interview where the informant did not consent to being recorded, notes were taken and later verified with the informant. When possible, additional data such as specific project documentation was collected.

Informants in this study came from technology-driven organizations in different industries and different hierarchical positions, thus providing varying perspectives on their service design practices. These considerations are regarded as helpful in mitigating the risks of potential biases where sampling is concerned

(Eisenhardt and Graebner, 2007). Data in this study was analyzed using an abductive approach (Alvesson and Sköldberg, 2009). Although I did not use pre-determined categories for coding, my analysis relied on frameworks and conceptualizations that were already established in the literature. Data was coded iteratively in two analysis cycles (Saldaña, 2015). In the first cycle, I used preliminary coding to detect reoccurring patterns. In the second cycle, the focus was on clustering the codes into categories and, by exploring their mutual relationships, and developing themes.

3.5. Research evaluation

Evaluating qualitative research is challenging, and not only because it is often viewed as fluffy compared to quantitative studies (Miles, Huberman and Saldaña, 2018). In this section I will refer to the criteria of validity and transferability, two criteria that are often considered fundamental for evaluation.

3.5.1. Validity

Whittemore et al. (2000) argue that establishing validity in qualitative research is challenging because of the inherent contradictions that require the researcher to show rigour, subjectivity and creativity in the research process. Validity represents how trustworthy the findings of the research endeavor are, and it can be demonstrated in several ways. Triangulation or acquiring multiple data sources (Creswell and Poth, 2018) is considered to be one such strategy for ensuring validity. Multiplicity of data sources can be related to multiple people, times or places of data collection (Flick, 2018). In Study 2 I relied extensively on triangulation as the study was designed to collect not only interview data but also data from observations and secondary sources, and at multiple points in time. Triangulation is also one of the major tenets of the case study method that build its credibility (Eisenhardt, 1989). Although interviews with informants were the main source of data collection in Study 3, I have, whenever possible, obtained other data sources. This included additional internal documentation and publicly available organizational information in the process of data interpretation. During the interviews in both empirical studies, I have tried to create a safe and reflective atmosphere so that informants felt validated in openly sharing their own experience and knowledge regarding the topic in question. Throughout the different stages of the research process in both of my studies, and in particular

during the data analysis, I have had an ongoing open discussion with my supervisors. Although I was the primary coder in both of the studies, I frequently discussed the coding process and the emergence of categories and themes with my supervisors and co-authors. Their feedback was then integrated into further iterations of coding and was consistent throughout the analysis process. Study 2 resulted in published paper that has gone through several rounds of peer-review. I have also sought out peer feedback at the department and during conference presentations as a form of credibility checking (Lincon and Guba, 1985), which was instrumental in pointing out inconsistencies, vagueness, or potential biases in the findings.

3.5.2. Transferability

The epistemological nature of qualitative research usually means that the type of generalizability of results that is expected in quantitative studies is unattainable (Guba and Lincon, 1994). In my studies I also do not claim such generalizability. Although Study 2 is a multiple longitudinal case study that could potentially provide grounds for inferring general conclusions about causal relationships (Leonard-Barton, 1990), and Study 3 is an interview study with a good theoretical sample (Eisenhardt and Graebner, 2007), both empirical studies are exploratory and context specific. They aim for in-depth understanding of service design in relation to different building blocks of service innovation. For that purpose, I have used a thick description of service design practices and technology-driven context idiosyncrasies as a way to advance knowledge on the intersection of service design and service innovation that requires further theoretical development (Antons and Breidbach, 2018).

3.5.3. Critical reflection

Working with qualitative approaches has been an ongoing challenge. Throughout the two empirical studies I have learned that qualitative approaches require both extensive discipline and creativity in planning, execution of data collection and data analysis. In my empirical studies I have relied on the coding protocols suggested by Saldaña (2015), which helped me in systematizing data, tracking the development of my interpretation and engaging in reflexive thinking. I have also engaged in iterative discussions with my co-authors during the analysis of my empirical work. Through these steps I wanted to be sure to minimize the

possibility of a confirmation bias in my interpretation (Graebner et al., 2012). In line with the epistemological stance of this thesis, I want to acknowledge that my previous experiences have shaped the interpretation of my work, but also that the knowledge construction in this thesis was a joint endeavor between myself as a researcher and the informants of my study situated in their organizational contexts. The case study approach and the qualitative interview study have enabled me to delve deep into the exploration of service design in technology-driven context, and to work with data that was nuanced and rich in order to complement and extend prior theoretical work. It also informed my conceptual work that was evolving in parallel as it helped me think about the abstract concepts in a more concrete way and thus contributed to the theorizing process.

Finally, I would like to address some ethical considerations. Although my studies did not involve vulnerable participants, which would have necessitated the ethical approval of my studies, I did use informed consent in all my studies before collecting data. Informed consent covered the verbal data that was disclosed during observations and/or interviews and any additional written material, such as internal reports, if informants decided to share it. Before starting with the data collection, I informed the study participants about the purpose of my study, the way I intended to use and store the data that I acquired, and asked permission for audio recording and note taking. I have also ensured that all the data will be anonymized. Only after informing them about the study and obtaining the consent did I start with the data collection.

4. Overview of the contributing papers

This chapter provides an overview of the individual research papers that contribute to this thesis.

4.1. Paper 1: Bridging design-driven and service innovation: Consonance and dissonance of meaning and value

The aim of this conceptual paper is to explore how the concept of meaning in design-driven innovation (Krippendorff, 2006, Verganti, 2008) and the concept of value from service-dominant logic (Vargo and Lusch, 2016) are interconnected. The two concepts seem to share common characteristics and conceptual grounds; however, their interrelation and commonalities have not been explored. In this paper we argue that this connection is necessary in order to understand the conceptual convergencies of service design as an interdisciplinary field that is rooted in both design and service research traditions. To do that, this paper explores two streams of literature: design-driven innovation, which relies on meaning as a central concept from human-centered design (Krippendorff, 2006, Verganti, 2008), and S-D logic, which focuses on interactive value creation as one of the main mechanisms for understanding how markets work (Vargo and Lusch, 2016). Both meaning and value are conceptual pillars in their respective disciplines. Despite recent developments in service design research, they have not been explored jointly although theoretical and conceptual integration of service innovation and service design is considered a priority (Ostrom et al., 2015). This paper was focused on explicating the concepts of meaning and value through a literature review to find common conceptual ground. The main purpose was to understand how the concepts connect. Based on the literature review the findings of this conceptual paper indicate five connectors where the two concepts converge. These connectors are: interaction, context, institutional generation, emergence, and idiosyncrasy (See Figure 2).

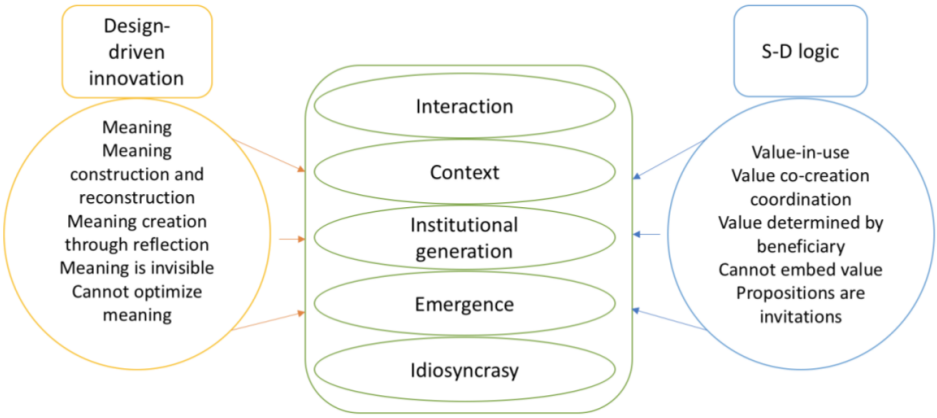


Figure 2: Connectors for meaning and value indicating interrelation

The connectors reveal that both meaning and value emerge during interaction (in use) and have idiosyncratic qualities but are situated in contexts that are governed by institutional norms, values, and rules. Thus, the findings in this paper suggest there is a conceptual convergence between the core concepts in human-centered design and service-dominant logic that is relevant to service innovation. This has implications for service design as a field that integrates design and service research perspectives but lacks a clearer conceptual position in service innovation, especially related to new value cocreation.

4.2. Paper 2: The role of meaning in service innovation: a conceptual exploration

The aim of this conceptual paper is to position the concept of meaning as relevant in understanding the conceptual building blocks of service innovation as defined in S-D logic; value propositions, resource integration and value cocreation. As such, it builds on the findings related to conceptual connectors from Paper 1. As a central concept of human-centered design, meaning explains how actors engage and interact and how design can support this (Krippendorff, 2006). Although S-D logic considers value to be meaning-laden and supports the notion that meaning is an important part of resource integration and institutional arrangements (Vargo and Lusch, 2008, Vargo et al., 2015), the meaning-making mechanisms that are relevant to service innovation in S-D logic are latent and lack thorough explication. To address this challenge, conceptual integration is undertaken (MacInnis, 2011), where the concept of meaning from human-centered design

research is positioned in relation to value propositions, resource integration and value cocreation to explicate its role in the service innovation process. Meaning is conceptualized (see Figure 3) relative to service innovation as having interpretative power and serving as a mechanism for resource interpretation (resources becoming), which enables resource integration and subsequent value cocreation.

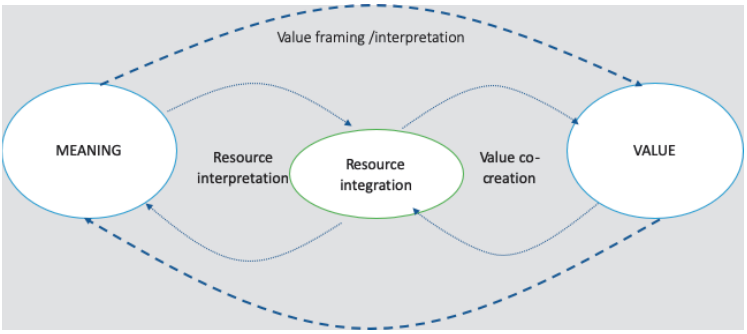


Figure 3: Conceptual dynamics of meaning and value

The paper contributes to S-D logic by extending the understanding of the building blocks of service innovation. By positioning and delineating the concept of meaning it explains how actors form intentions to engage in the value cocreation process through resource interpretation. This is an important link which is currently missing from the assumptions surrounding resource integration and value cocreation in S-D logic. This has implications for service innovation since it provides an interpretative and human-centered perspective that strengthens the phenomenological and experiential aspects of new value cocreation (see Figure 4).

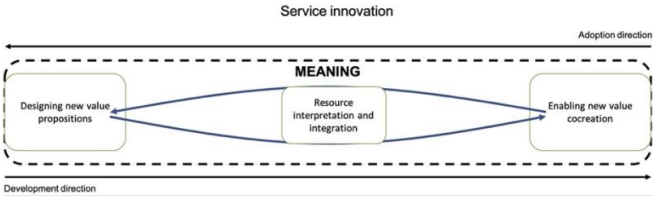


Figure 4: Understanding meaning in service innovation

Following the conceptual exploration, future research directions are provided with the aim of inspiring further research at the intersection of service design and service innovation. Future research directions present possible research questions about meaning and each of the building blocks of service innovation.

4.3. Paper 3: Service design as an innovation approach in technology startups: a longitudinal multiple case study

The aim of this empirical paper is to understand how service design facilitates service innovation in technology startups. Technology startups are one of the most important actors in the innovation ecosystem, but they suffer high failure rates (Giardino et al., 2014). Due to their restrictive value focus in different stages of their life cycles, service design can help in creating meaningful and a value-centric service offerings (Klotins et al., 2019, Patrício et al., 2011). As little is known about how service design facilitates innovation in technology startups and what its outcomes are, this study undertakes a longitudinal multiple case study methodology and investigates 5 technology startups before, during, and after their participation in service innovation projects in which a service design approach was used. This study frames value propositions as a primary mechanism of service innovation (Skålén et al., 2015). The results reveal a focus shift in creating value propositions facilitated by service design during the innovation process. They also reveal how service design was embedded and contextualize the potential for leveraging service design opportunities relative to the startups' life cycle stages. A summary of findings is presented in Figure 5.

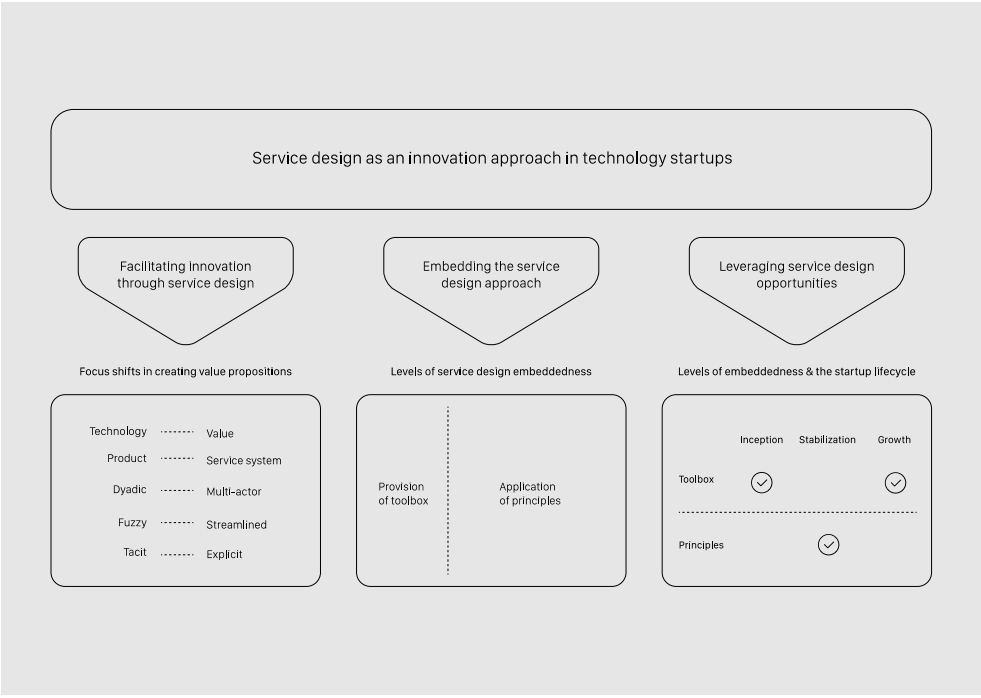


Figure 5: Framing service design as an innovation approach in technology startups

This study contributes to service innovation literature by shedding light on value propositions as mechanisms for service innovation (Skålén et al., 2015, Payne et al., 2017). It shows how service design can shift the value proposition from product-centric to value-based in the following ways: from technology to value considerations, from dyadic to multi-actor perspectives, from product to service systems, from fuzzy to a streamlined process and from tacit to explicit representation. This shift enabled startups to understand the different drivers of their value-creating opportunities. However, the findings also reveal nuances that are particular to the startup context. The study suggests that the shift in creating value propositions facilitated by service design may still result in different levels of service design embeddedness, one narrower (provision of toolbox) and another broader (application of principles) in scope. The provision of a toolbox reflects the understanding that the service design approach is a method with a checklist type of process that primarily requires knowledge about the correct use of tools. Application of principles reflects the understanding that service design brings a human-centered, multi-actor and value-based focus to the innovation process, and

that the application of these principles, rather than the tools themselves, represents the value of the service design approach. Finally, the findings indicate that the level of embeddedness might be connected to a startup's life cycle stage, which has implications for leveraging service design opportunities. Namely, the "toolbox" levels correspond to the inception and growth stage, while the "principles" level corresponds to the stabilization stage. In practical terms this indicates that timing (or the startup's life cycle stage) is important if service design is to be introduced as part of the innovation process because it might have consequences for its expected scope and impact.

4.4. Paper 4: We're in, now what? Integrating service design for sustained service innovation

The aim of this paper is to understand how established innovative organizations use service design to build capabilities for sustained service innovation. Innovative organizations increasingly seek to use service design to support service innovation and organizational transformation (Kurtmollaiev et al., 2018), however little is known how it helps them in building lasting innovation capabilities for continuous adaptation (Malmberg and Wetter-Edman, 2016). Additionally, the success of the service design integration is often reported in relation to external consultancy projects, so its ongoing organizational contribution to innovation is unclear (Holmlid and Malmberg, 2018). In this study we undertake a qualitative interview study with expert service designers in established organizations that have been using service design internally to innovate on a continuous basis. The aim is to understand how service design enables sustained service innovation and for that purpose dynamic capabilities framework is used as an analytical lens (Teece, 2007).

The findings reveal eight service design-related practices that underpin the sensing, seizing, and transforming capability in enabling sustained service innovation (Figure 6). These are: (I) Articulating rich knowledge about the service system, (II) Reframing the problem-solution space, (III) Interpreting future service system needs, (IV) Generating human-centric service concepts, (V) Embedding systems perspective during service development, (VI) Infrastructuring service design for implementation, (VII) Activating the base for service design integration and (VIII) Cocreating design culture. The findings also point to tensions that hinder organizational integration of service design such as misalignment of design practices with organizational culture and knowledge

gatekeeping. Results reveal that service design practices contribute dominantly the sensing capability. This has implications for how service design integration is supported along the innovation process including implementation activities and managerial support. This study contributes to scarce literature on service design and capability building for innovation.

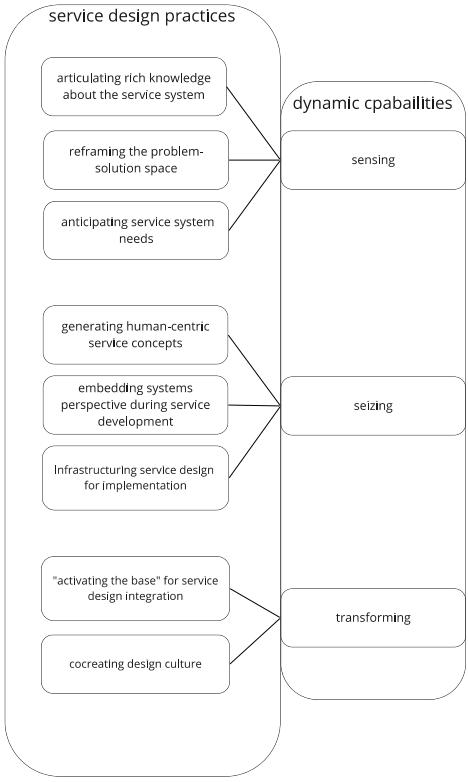


Figure 3: Framework of service design practices underpinning dynamic capabilities

5. Discussion

The discussion section consolidates the findings from the individual studies in this thesis. In this section I will discuss the general contributions of this thesis, explore implications for research and practice and finish with the limitations.

5.1. Revisiting research challenges

As markets are becoming complex, interconnected, and infused with technology, many organizations are turning to service innovation. As a multidimensional phenomenon involving value cocreation among multiple system actors, service innovation is seen as a research priority (Ostrom et al., 2015) and service design as an interdisciplinary approach to innovation that can facilitate it on different system levels (Patrício et al., 2011) while also enabling system transformations (Kurtmollaiev et al., 2018, Vink et al., 2021). In this thesis both service design and service innovation take S-D logic as a unifying theoretical perspective (Wetter-Edman et al., 2014), providing an analytical lens through which to understand how new value is interactively formed. However, due to the interdisciplinary nature of service design and its conceptual trajectory in the field of service research (Prestes Joly et al., 2019) its role is often based on the underlying assumptions that direct service innovation possibilities. In Chapter 2 I have pointed to the lack of integration at the intersection of service design and service innovation (Antons and Breidbach, 2018, Yu, 2017). While research suggests that building on service design's multidisciplinarity can strengthen its impact in service innovation (Ostrom et al., 2015, Patrício et al., 2018), this can also create confusion in conceptual positioning, resulting in contradictory research agendas that can limit its theoretical development. Integrative research that seeks to explicate the multidisciplinary dimension of service design is scarce (Prestes Joly et al., 2019, Yu, 2020). Thus, while some researchers problematize service design as an all-encompassing innovation concept, suggesting that it should be narrowed down to a development process with a design perspective (Gustafsson et al., 2020), others view it as insufficiently broad and call for its wider inclusion in understanding change on a systemic level (Koskela-Huotari et al, 2021, Sangiorgi, 2011). These inconsistencies also reflect the ongoing challenges in defining service innovation, especially in terms of how change emerges and who gets to decide whether new value has been created (Snyder et al., 2016). Additionally, I have problematized the assumption related to the

practical nature of service design in service research that neglects rich, albeit multifaceted theoretical traditions from design research. I have particularly emphasized the foundational work from Krippendroff (2006) and his concept of meaning. By integrating the concept of meaning in relation to value cocreation (Paper 1 and Paper 2) this thesis extends the understanding of interpretative and experiential aspects of service innovation (Helkulla et al., 2018, Rubalcaba et al., 2012). This has important theoretical implications for service design and service innovation. It explains why service design is relevant to service innovation beyond its methods and tools, and how meaning can enrich the conceptualization of new value cocreation through a uniquely design-oriented perspective founded in design research. Thus, meaning can be understood as a design source for value-based service innovation.

Despite the growing research interest in service design, its stronger theoretical and conceptual integration with service innovation is still under development (Antons and Breidbach, 2018). As this thesis is intersecting service design and service innovation, my studies addressed different aspects of service innovation. I conceptualize service innovation within a synthesis perspective using S-D logic, thus the contributions connect to a value-based view of service innovation (Helkkula et al., 2018) and its two mechanisms: value propositions (Skålén et al., 2015) and resource integration (Lusch and Nambisan, 2015). The conceptual study (Paper 1 and Paper 2) is concerned with meaning-related aspects in new value cocreation as a way of theorizing about service innovation. The empirical studies (Paper 3 and Paper 4) focus on service design relating to changes in value proposition for new value cocreation, and changes in resource integrating practices enabling organizational capabilities for service innovation. Based on the findings, in the next paragraphs I will first articulate and discuss the role of service design in service innovation. Next, I will discuss meaning as a design source of service innovation and consider the implications of its different use.

5.2. Understanding the role of service design in service innovation

In service research, service design is mainly represented as an approach to service innovation and is seen as its main driver (Grenha Teixeira et al., 2017, Maffei, Mager and Sangiorgi, 2005, Patrício et al., 2011). This is grounded in its historical link to new service development and quality management, as service research acknowledged the necessity for systemic and experience-centric service

development, implementation, and control (Alam and Perry, 2002, Edvardsson and Olson, 1996, Zomerdijk and Voss, 2010). Based on the theoretical background presented in Chapter 2 and the empirical studies (Paper 3 and Paper 4), in this section I will articulate two ways of understanding the roles of service design in service innovation: as creative exploration, and as ongoing transformation. Figure 7 shows how both roles connect to archetypal dimensions of service innovation (Helkkula et al., 2018).

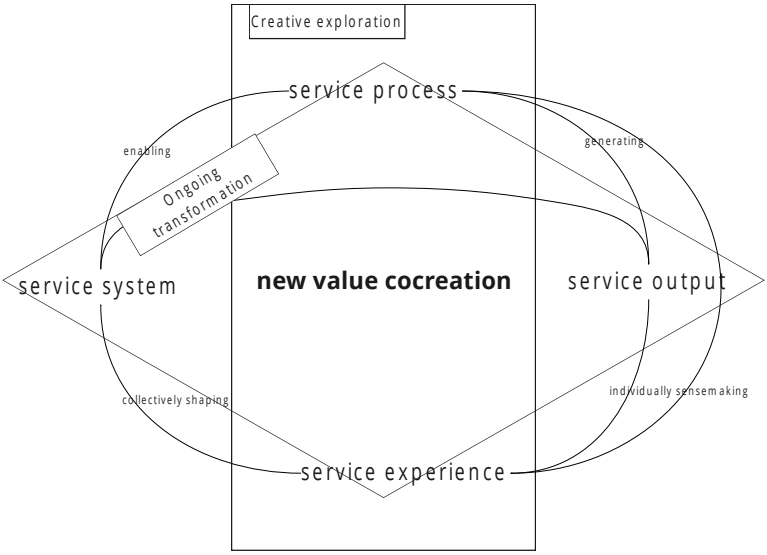


Figure 4: The role of service design in value-based service innovation framework, adapted from Helkkula et al. (2018 p.293)

5.2.1. Service design as creative exploration

Recent research has problematized the perspective in which service design is viewed as an early phase in the NSD process, arguing that it should be integrated during the development and implementation phases to pivot from the value cocreation perspective (Yu and Sangiorgi, 2018b). Thus, service design as an instrument of creative exploration embraces the underpinnings of S-D logic in understanding that service design outcomes are always cocreated with resource-integrating actors (Holmlid et al., 2017). While it extends the service design role in the development process, the dominant focus of service innovation is the

process of designing for new value cocreation on a level between service providers and focal beneficiaries. As it can be seen in the Figure 7 it is primarily connected to the process and experience archetype of service innovation (Helkkula et al., 2018).

In creative exploration the object of service design is focused on designing a service concept that supports a new value proposition. Value propositions are understood as key mechanisms of service innovation that consist of configurations of resources and practices (Skålén et al., 2015). In other words, organizations engage in this type of service design when the focus of their innovation is to experiment with an emerging idea to see how it might translate into feasible service concepts within their current strategic positioning. The value of service design is in the process of developing something new and potentially value creating for the customer, while the implementation is often not in the focus. Thus, as shown in Paper 3, the motivation for organizations to introduce service design as an approach is to experience a different – and possibly more creative – way of doing their innovation work with a value cocreating potential. This role of service design is common in projects, as temporary structures (Holmlid, 2015), with external service design experts, although it can be used in organizations with internally embedded service design practices as well. The project brings together organizations as clients, usually without design expertise, and service designers as external experts, meaning that service design facilitates innovation work within temporary structures that can direct innovation processes and outcomes based on collaborative relationships. This finding is in line with Yu and Sangiorgi (2018a) that show how different client-designer relationships facilitate different outcomes.

Organizations also act as challenge owners that decide how the innovation challenge is defined, which can range from more to less open. Paper 3 shows how service design facilitates service innovation through creative exploration in these temporary structures by focusing on the service design enabled changes in designing new value propositions that are strongly related to technology. This paper also provides perspective on startups as organizations that, due to a strong focus on technology, can often neglect a user-centric perspective as Paradkar, Knight and Hansen, (2015) show. The paper also shows how the starting point of the innovation challenge differed between organizational members directing the possibilities of designing a new value proposition, which at the end of the project was decisive in its implementation. Commonalities were observed in designing

new value propositions, mostly related to reconfiguration of technological resources, and redefining the value proposition from the multi-actor and systemic perspectives.

Thus, in creative exploration the service design process plays a major role in designing new value propositions, since it is framed as a relatively structured process employing specific methods and tools, moving through different stages of design. In the empirical material at hand, it usually follows the double diamond framework (British Design Council, 2005), focusing on methods and tools that allow divergence and convergence between formulating problems, generating and testing ideas, and creating the final solution. The process is infused with human-centric, participatory, and holistic principles (Blomkvist, Holmlid and Segelström, 2010, Holmlid, 2012) that engage project participants in collaborative techniques for exploring the service system and visualizing current and future service components. As such, organizational members that are part of temporary structures within which service design is practiced, also assume a designer role as they are proactively included in design activities that aim to empathize with customers and users, ideate novel service concepts and prototype solutions.

Service design in creative exploration is usually a novelty for organizational members. The motivation to use service design is framed within a "thinking outside the box" assumption. This assumption is common in design thinking (Brown 2008). The notion that everyone can design, that it can be achieved through a hands-on approach and that it can be used for future service design challenges underlines this connection between innovation and organizational learning. However, this can put too much focus on tools and methods themselves rather their purpose, and influence future resource integration opportunities and development of capabilities to innovate. As Paper 3 shows, the organizations in the study did exhibit learning outcomes connected to how they made sense of service design during the project collaboration. Apart from collaboratively designing a new value proposition, organizational members learned how they could innovate with a greater focus on users and other system actors, how they could understand complexities of different values in the service system and how they could reinterpret technology through value in use. While both Paper 3 and Paper 4 show that learning is indeed one of the organizational outcomes of facilitating service innovation through service design, it also shows variety in interpretation. For example, the toolbox and principles outlook on service design

embeddedness reported in Paper 3 reflects how organizational members perceive both meaning and value of service design during the innovation process.

Some authors have argued in favor of service design projects enabling development of a user-centric mindset and fostering innovation capabilities (Steen, Manschot and de Koonig, 2011). However, in the case of creative exploration, Paper 3 shows that this might be difficult to achieve. Åman, Andersson and Hobday (2017) differentiate between learning to design and learning through design in discussing the scope of design knowledge. In Paper 3 the value of service design knowledge as a core operant resource for service innovation was dominantly understood as knowing how to use specific tools in specific design stages. Reflection on what kind of knowing these tools can enable was not as dominant and was possibly a result of the way project was structured. As Paper 3 shows with toolbox and principles levels of embeddedness, innovation capabilities are not probable organizational outcomes of such projects. The inability of organizations to embed service design in their ongoing innovation activities after projects finish is also highlighted in earlier research (see e.g. Bailey, 2012). Paper 3 shows that a toolbox orientation will facilitate some understanding of service design's potential but will be considered difficult to implement in ongoing innovation activities due to a perceived lack of expert design knowledge. Thus, in creative exploration, learning is done on a non-continuous basis with external sources. While this can enhance creative efforts in designing value propositions, it also implies that expert service designers might have limited influence in the implementation phase, which can in turn hinder the understanding of service design as being strategically important. Similar findings were confirmed by Overkamp, 2019.

Thus, the intent of service design as creative exploration is the potential change in firm and customer value creation achieved by designing new and improved value propositions. While designing new value proposition can include some systemic considerations, lack of organizational structures that can proactively support meaningful embeddedness of service design could hinder its potential for a more transformative change. Service design as creative exploration can be an important way for organizations to experience and learn about the value of service design in service innovation. However, organizations should also be mindful of the assumption that exposure to that experience will seamlessly translate to organizational capability to innovate.

5.2.2. Service design as ongoing transformation

Recent service design research has increasingly focused on understanding how service design facilitates service system transformations (see e.g. Fisk et al., 2018, Koskela-Huotari et al., 2021, Kurtmollaiev et al., 2018, Vink et al., 2021). Koskela-Huotari et al. (2021) see service design as a way to intentionally influence transformation of firms, industries and societies by collectively and proactively involving actors across different system levels in imagining desirable futures. They also argue that transformation can be achieved only by changed practices that become institutionalized. This indicates that organizations increasingly need to consider their positions in a wider network of actors for which they need both adaptive and transformative capabilities (Rodrigues, 2020). Although change can ensue independent of innovation efforts, service innovation represents a constitutive element of service system transformation, including organizational level change (Crossan and Apaydin, 2010). Thus, service design as an ongoing transformation and its role in service innovation necessitates capability development. As Paper 4 shows, continuous embeddedness of service design practices enables development of organizational capabilities for sustained service innovation. This enables organizations to proactively shape the innovation processes and value cocreating outcomes based on understanding service system opportunities. Thus, as Figure 7 shows service design as an ongoing transformation encompasses all service innovation archetypes (Helkkula et al., 2018).

Paper 4 argues that ongoing integration of service design practices is necessary for building capabilities for service innovation. However, organizations need to create conditions under which such organization-level capabilities can be fostered (Karpen et al., 2017, Malmberg, 2017). This, on the other hand, involves ongoing institutional effort (Kurtmollaiev et al., 2018). Paper 4 indicates that internal embeddedness of service design is necessary for developing organizational capabilities for service innovation. In service design as ongoing transformation organizational motivation for embedding service design goes beyond experimenting with creative methods to address a specific service challenge. It focuses on understanding how collaborative, human-centered and systemic practices of service design can facilitate collaboration among different system actors for new value cocreation. Thus, organizations increasingly hire or train service design experts that work in internal cross-functional teams, usually on project-based innovation challenges. While these projects can also be considered

as temporary structures, their situatedness in an organizational context is stronger as they constitute the ongoing work. This influences how service design is practiced, as Paper 4 indicates. Thus, service design practices are contingent to organizational level innovation conditions that reflect how projects are framed.

While projects in creative exploration tend to follow some simplified process model framework and use a repertoire of established tools, this is rarely the case with service design as an ongoing transformation. As Paper 4 shows, methods and tools are modified according to the context of the project. Thus, service design facilitates service innovation by employing continuous practices that enable collaborative learning about and with the service system actors, reframing innovation challenges and supporting management decisions across the stages of development and implementation. These practices can then support capability development relating to sensing, seizing, and transforming organizational operant resources. Thus, the motivation for organizations to embed service design stems from their understanding that it can help in building capabilities for sustained innovation. This represents their primary reason for engaging with service design as an ongoing transformation. However, building design capability is a complex and lengthy process since capabilities represent the interplay between design practices and organizations as enabling or restraining structures. This requires a strategic commitment and design leadership. This is in line with Micheli et al., 2018, that call for elevating the design function in organizations as a precondition to capability development. As Paper 4 shows, legitimizing the role of service design for innovation is important to give visibility and validity to the work of service design experts. While service design experts will usually self-initiate collaborative design work with cross-functional team members who might not understand the potential of service design for innovation, Paper 4 argues that this bottom-up approach is to some extent limited in its impact. While it helps in creating a shared understanding of design practices, values, and language for smoother collaboration, it might not converge into an organization-wide design culture. Data in Paper 4 indicate that support from the highest organizational level is crucial in normalizing strategic role of service design in innovation work and activating the base for continuous service innovation capability development.

Thus, the intent of service design as an ongoing transformation is the development of capabilities that can enable sustained service innovation. Service design is seen as an ongoing process of continuous value cocreation. It underpins capability

development supporting changes in resource integration that drive service innovation. Thus in service design as an ongoing transformation learning is preoccupied both with the design process and knowledge that results from it. Learning through design (Åman, Andersson and Hobday, 2017) is seen as a key knowledge integration process and Paper 4 shows that this type of learning is especially relevant in supporting organizational capabilities for sensing market opportunities, where service design helps to articulate knowledge about service system actors, reframing the problem/solution space and interpreting future service system needs. In this role service design can be understood as a way of knowing that is embedded in the context of the organization which can be enabling or restrictive. Thus, for supporting service design as an ongoing transformation, the alignment of service design with exiting organizational structures and legacies matter.

5.3. Meaning as a design source of service innovation

In Chapter 2 I discussed positioning of service design in service research as an innovation approach that combines service perspective and design practice (Patrício et al., 2011, Prestes Joly, 2019). Value-based service innovation sees new value cocreation as a source of service innovation (Vargo and Lusch, 2008) and service design as a practice that can facilitate this emergence (Yu and Sangiorgi, 2018b). Thus the design part of service design is often viewed dominantly through practical perspective, neglecting multifaceted design research traditions (see e.g. Holmlid, 2009, Kimbell, 2011, Sangiorgi and Prendiville, 2014). In this thesis in Paper 1 and Paper 2 I propose that the concept of meaning (Krippendorff, 2006) from design research should be viewed as an integral part of explaining the emergence of new value. Thus, I position meaning as a design source of service innovation. Krippendorff (2006) sees meaning as essential in understanding human interaction with design artifacts on individual, social and cultural level. Based on centrality of meaning for human interaction, meaning as a design source of service innovation can conceptually inform new value cocreation with design research perspective. Interpretative dimension of design and its ability to work with meaning has been previously discussed in design research (Jahnke, 2013, Kimbell, 2011, Wetter-Edman, 2014), however, the stronger conceptual integration of meaning surrounding service design and service innovation is still missing. Paper 1 and Paper 2 extensively discuss how meaning and value converge on important conceptual trajectories, but still entail

different purposes. Viewing meaning as a design source of service innovation articulates design perspective of service innovation and provides a theory-driven insight into its human-centered principles.

Meaning as a design source of innovation can inform both roles of service design that I have discussed before. For creative exploration, where process and experience dimensions of service innovation are dominant, meaning of artifacts in use (Krippendorff, 2006) can guide value cocreation by focusing on potential value in experience (Helkkula et al., 2012) or value in use (Vargo and Lusch, 2008) integrated in value propositions. For ongoing transformation, a systemic level of meaning connected to language and ecology of artifacts (Krippendorff, 2006) can guide value cocreation by focusing on potential value in social context (Edvardsson et al., 2011) integrated in value propositions.

5.4. Implications for research and practice

In this thesis I articulate two roles of service design in service innovation and position meaning as a design source of service innovation. Although I explicate two different roles of service design in service innovation I see the difference between them as a matter of foci, rather than clear boundaries. As such, they can serve as a starting point for understanding what the purpose of service design in facilitating service innovation is, how is service design used and how it contributes to the organizational capability to innovate.

First, service design as creative exploration is focused on designing new value propositions within temporary structures that provide temporary experiential learning. Service design as an ongoing transformation is focused on continuous integration of service design practices within permanent structures that foster service innovation though organizational capability building. Understanding different service design roles in service innovation contributes to service innovation and service design literature by reducing the ambiguity that surrounds service design as a multidisciplinary concept (Prestes Joly, 2019, Yu, 2020). It connects them to value-based innovation framework and positions them according to dominant archetypes (Helkkula et al., 2018). Thus, different foci of service design roles allow focusing on different service innovation mechanisms. Service design can facilitate value-centric change in new value propositions and this contribution is discussed in Paper 3. However, it can also facilitate novel

forms of resource integration through development of innovation capabilities. This contribution is discussed in Paper 4. For service designers these contributions can be relevant to understand the scope of their engagement and potential impact to service innovation. Creative exploration can enable service designers to focus on designing new value propositions within the temporary boundaries of a project. While this may have limited impact on the implementation part of service innovation, it may still allow organizational actors and service designers to experience service design principles and tools through learning about design. Ongoing transformation can enable service designers to focus on continuous alignment of service design practices within organizational context and foster service innovation through capability building. Since organizational conditions may influence the quality of alignment, service designers should proactively address organizational design legacies (Junginger, 2015) that could either foster or hinder service innovation capabilities.

Second, service designers increasingly design for complex service systems. This includes dealing with many different service innovation dimensions and understanding how they influence value cocreation. In dealing with this complexity, understanding meaning as a design source of innovation can play an important role. Thus Paper 1 and Paper 2 can serve as a starting point for this consideration. For example, take a self-driving car. It is a complex artifact because its technological challenges are embedded within current transportation infrastructure that raises many legal and wider social questions. On a surface level, or what Krippendorff (2006) would consider a first order understanding, a self-driving car may seem like the same artifact as a traditional car because it still provides transportation from point A to point B, albeit with some much-improved functionalities. Thus, the existing understanding is influenced by the existing mental models of what a car means, how it is used, and how it connects to other artifacts in the wider system of transportation. However, if we take away the use connected to driving, what kind of experience is left? What does it mean to drive a self-driving car? Will people drive self-driving cars, or just travel in them? Will the activity of "driving a car" as we know it become meaningless? The technologically enabled autonomy of a car profoundly changes the meaning of its use. It influences actors' meaning making across different levels with consequences related to interaction with transportation infrastructure, changed behavior and consumption patterns both individually and socially, and has consequences on policy making. Thus, emphasizing meaning-making

mechanisms in new value cocreation allows service designers and others system actors to become more attentive to the interconnected elements that a use situation entails and the way it could change. The focus on meaning is complementary to value cocreation, but what meaning enables is knowledge of where to look for emerging change. It provides a sense of direction for service designers when dealing with ambiguous and complex problems by directing them to look for meaning of artifacts in use, language, artifacts' lifecycle, or their ecology (Krippendorff, 2006). Literature has established that designers possess interpretative capacity that is often situated in aesthetic dimensions of their practice (Jahnke, 2013). In line with that, I argue that this dimension should be understood and nurtured in service design for service innovation.

5.5. Limitations

This thesis has several limitations. First, the empirical work in this thesis is situated within the context of innovative organizations with technology-driven focus. In Study 2 I report on service design projects in technology startups, while in Study 3 I report on service design integration in large innovative organizations with a strong technology focus. Both types of organizations are strongly oriented towards innovation, and I would contend that innovativeness represents an essential part of their organizational culture and identity. For example, organizational cultures that encourage innovation, experimentation and entrepreneurial action might manage service innovation, including service design, differently from those who do not encourage it. Additionally, the organizations in Study 3 were all successful large organizations with a historical track record in successful technology and product innovation. Thus, the articulation of two service design roles in service innovation could reflect an ideal case. Additionally, empirical studies were situated and conducted with informants working in international, Western organizations. This means that my informants could have manifested implicit biases about the meaning of innovation and organizational success pertaining to developed capitalist societies. Therefore the results of the empirical studies are not transferable across contexts.

Second, although the context of innovative organizations represents a common denominator in this thesis, there are some methodological limitations that I would like to discuss. Study 2 was conducted on a longitudinal basis with startups in Portugal. During the data collection I used the English language to communicate

with my informants and participate in observation. However, with some informants, the language barrier was prominent, which affected how they could express their views, especially during the semi-structured interviews. This has affected the semantic richness of the data that I could obtain. Additionally, data collection involving observations of service design work depended on the schedule set by the project team members and startups. Although investigating collaboration was not the purpose of this study, there were differences in collaboration frequency among teams in service design projects that could influence the results. Study 3 was conducted with service design experts across different organizations working with technology-enabled service innovation. Selecting and accessing organizations and informants that have been internally embedding service design was challenging. The difficulty in accessing informants resulted in a longer than planed data collection period, which also affected my qualitative analysis. I concur with Saldaña (2015) who sees qualitative analysis as emerging between the informants and the researcher as the research progresses. I have tried to remain reflective about the data collection and preliminary analysis that was already done before continuing data collection with subsequent informants. However, as this progress was non-linear, it did affect the dynamics of data analysis as well as the time balance between the first and the second cycle coding. While in Study 2 I have had the opportunity to participate in observations and conduct interviews face to face, this was not the case with all informants in Study 3. Out of 21 interviews, four were conducted over the phone, and three were in person but outside the informant's workspace. Although I designed Study 3 with interviews as the primary data collection instrument, the lack of situatedness in the informants' contexts and/or inability to physically interact might have affected their level of comfort and trust with me as a researcher, and consequently, the data they were willing to share.

Finally, this dissertation is concerned with multidisciplinary concepts of service design and service innovation. My main theoretical lens in understanding service innovation and service design has been S-D logic. This means that the starting point in understanding service design comes from services marketing and management literature that is aligned with my educational background in business. Although in Chapter 2 I discuss service design from design research perspectives and use design concepts in my conceptual study, the lack of design experience at the outset of my PhD also influenced my research work. Ongoing divergence and convergence between multidisciplinary theories and concepts

from service and design research in this thesis are reflected in rather visible divisions between the empirical and conceptual papers. However, I see this interplay between breadth and depth in multidisciplinary topics as a reality of this type of research.

6. Conclusion

The purpose of this thesis was to understand the role of service design in service innovation for two reasons. One was to articulate its position within value-based dimensions of service innovation. Two was to understand a service design mechanism in new value cocreation. The results of these thesis are based on three studies, one conceptual and two empirical.

This thesis positions service innovation as a value-based phenomenon grounded in S-D logic, understanding it through value propositions, resource integration and value cocreation. S-D logic has served as an overarching analytical lens in contextualizing the findings. Additionally this thesis integrates various literature from service design, human-centered design, service research and organizational capabilities to investigate different facets of service design and service innovation. Thus, the contributions of this thesis address the research call for conceptual and theoretical development on the intersection of service design and service innovation (Antons and Breidbach, 2018).

This thesis makes the following contributions. First, this thesis articulates two different roles of service design based on the material from two empirical studies. These are service design as a creative exploration and service design as an ongoing transformation. While the former explicates the role of service design projects in designing new value propositions and connects to process and experiential dimensions of service innovation, the later focuses on continuous organizational integration of service design practices that support organizational capabilities for service innovation, encompassing entire value-based service innovation. Thus, by articulating the roles of service design in service innovation, this thesis anchors diverging conceptual lens concerning the purpose of service design in service innovation (see e.g. Gustafsson et al., 2020, Vink et al., 2021). Service design as creative experimentation can help organizations facilitate change in their value propositions for new value cocreation through employment of service design methods and tools. On the other hand, service designers have an opportunity to facilitate this process while understanding the constraints related to organizational lifecycle. However, knowing how to use service design methods and tools might not be enough if the organizational goal is to developing innovation capabilities through service design. Integrating service design as an ongoing transformation, can help organizations continuously innovate by supporting internal service design practices. However, using service design for

development of innovation capabilities is a lengthy and complex process that requires organizational-wide alignment. Thus, service designers should consider how design leadership skills and roles can assist with that.

Second, this thesis articulates meaning as a design source of service innovation and positions it as a mechanism of understanding new value cocreation. Following conceptual integration of meaning (Krippendorff, 2006) and value cocreation (Vargo and Lusch, 2016) presented in the conceptual study, this thesis provides a theoretical underpinning of service design in service innovation. Although interpretative dimension of service design is not unknown (Kimbell, 2011, Wetter-Edman, 2014), this thesis argues for its stronger inclusion in service innovation because meaning can enrich how new value cocreation is understood. Finally, looking for clear boundaries in a research field that is still maturing is difficult and at this point I would argue that such attempts might even be reductive. There is still much we could learn about service design and service innovation. In particular, I see the research on service design capabilities as a fruitful research direction that is growing, and to which this thesis also contributes to. Service innovation is a complex phenomenon and dealing with complexity should be understood as a collaborative effort. From an organizational perspective, future research could investigate capabilities related to organizational learning as well as knowledge management perspectives. This can provide answer to questions of what constitutes organizational service design knowledge and how is service design knowledge integrated on different organizational levels, from cross-functional teams to organization wide structures.

Additionally, researching meaning as a design source of service innovation is still nascent. However, integrating meaning more explicitly in service deign practices for service innovation can become a fruitful way to explore individual and collective sensemaking within the experiential and systemic archetype of service innovation (Helkkula et al., 2018)

Coda: a personal reflection

In between. That would be the aggregate theme that I would assign to my dissertation and the journey that led to it. In between service and design, innovation and invention, empirical and conceptual, Portugal and Sweden, social sciences and humanities, value and meaning, a marketer and a musicologist. The last in between represents the dialectic of my professional identity. As I received my education from both business school and music academy, this dialectic was instrumental for this dissertation in all its complexity. However, until very recently I saw my dissertation building solely on my marketer identity. The reason why I got interested in service design was because of marketing. I saw service design as the missing link for most of what marketing was promising, but was rarely delivering, form customer orientation to sustainable development. And I really loved marketing as I have been teaching it for couple of years before starting my PhD. Practical marketing is very much concerned with "how" but rarely questions "why", and in business school the focus on "practical" is usually dominant. Marketing pillars about satisfying the customer while delivering a shareholder value is not something that gets questioned a lot. At least from my experience. However, marketing is probably the most misunderstood organizational function, and at that time I also believed that the role of marketing in innovation was massively undervalued. So when I first started learning about service design it felt like a world of opportunities has opened. At first, like with most early-stage research, concept often seem to fit perfectly on a surface level. It is only when you give it a closer look that you begin to see different patterns, textures, cervices, and holes that makes things the way they are.

My research in between service design and service innovation is in many ways trying to make sense of the boundaries surrounding both concepts. I also see it as an attempt to argue in favor of service design as a way to innovate service (in its broadest sense) because, under the right conditions, it allows complexity to be seen as a resource, rather than a constraint. I believe that this position is inherently inclusive. And I have come to a personal conclusion that while change could result in competence and value destruction, fostering inclusivity through collaboration can act as a turning point in how organizations, individuals and societies deal with change, and how they eventually innovate. To me, that is the value of (service) design.

Looking back to my in between identity, I realized one thing. One of the books that made a profound impact on me during my masters was Kuhn's "The structure of scientific revolutions". It was part of the reading list for my Systematic musicology course, where I vaguely remember discussing shifts in understanding and interpreting meaning in music under different music semiology paradigms. It has only occurred to me recently that both of my in between identities have found their way into my dissertation.

So rather than an end, I see my dissertation as the beginning. The beginning informed by learning and reflection about what it means to study something, making choices how to do it, and embracing the consequences of those choices. Although the *in between* was not the most comfortable position, in the end I appreciate its richness, and wonder whether for me it could have been any other way.

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